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Chiral Amplification Self-Training: An Exact Asymmetric-Catalysis Theory of Agreement-Filtered Pseudo-Labeling

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ABSTRACT Pseudo-labeling methods that keep only the unlabeled points on which several classifiers agree are among the most widely used tools of semi-supervised learning, yet, to our knowledge, the purity of the labels they produce has not been predicted from first principles. We show that this purity obeys, exactly, the non-linear-effect law of asymmetric catalysis recognized by the 2026 Nobel Prize in Chemistry. Mapping correct and wrong predictions to two enantiomers, a pair of agreeing classifiers to a homochiral catalyst and a disagreeing pair to a heterochiral "meso" complex, we prove that the label excess of agreement-filtered pseudo-labels equals Kagan's ML_2 expression with the meso reactivity equal to the rate at which disagreements are accepted. For independent members this reduces to a tanh law, $\varepsilon_{\mathrm{out}} = \tanh(M \operatorname{artanh} \varepsilon)$, and for correlated, multi-class committees we derive a Chiral Amplification Law (CAL) based on an effective committee size. A mean-field model of repeated self-training, analogous to Soai's autocatalysis, exhibits a pitchfork bifurcation at unit "chiral susceptibility" and predicts that amplification requires committee diversity. We further show that the out-of-fold accuracy estimates used inside self-training are leaked by the pseudo-labels themselves and propose fold-specific pseudo-label chains that remove the leak. These results yield CAST, a self-training algorithm that selects its committee by maximizing the predicted noise-adjusted yield and decides safely when to stop. On 34 public datasets, CAL predicts the precision of 20,400 real committees with a mean absolute error of 0.069 (0.029 with known statistics). CAST attains the highest mean accuracy, macro-F1 and average rank among 9 methods at both 10% and 20% label ratios, while classical self-labeling baselines frequently degrade accuracy.

INDEX TERMS Asymmetric catalysis, committee-based learning, ensemble diversity, non-linear effect, pseudo-labeling, self-training, semi-supervised learning, tanh law.

I. INTRODUCTION

LABELED data remain the bottleneck of supervised learning in medicine, industry and science, while unlabeled data are abundant. Semi-supervised learning (SSL) attempts to exploit the latter [1], [2], and its oldest and still most widespread family is self-labeling: a model trained on the few labeled examples assigns pseudo-labels to unlabeled examples, which are then added to the training set [3]–[6]. Because a wrong pseudo-label is learned as confidently as a correct one, the central difficulty of self-labeling is confirmation bias: errors feed on themselves [7]. The most popular remedy is to accept a pseudo-label only when several classifiers agree on it, as in co-training [8], tri-training [9], democratic co-learning [10], co-forest [11] and co-teaching [12].

Agreement filtering works remarkably well in practice, but it is used almost entirely heuristically. Practitioners choose the number of classifiers, which classifiers, and whether to require unanimity or a majority by trial and error. There is no formula that answers simple quantitative questions such as: if two classifiers are each 75% accurate and their errors are moderately correlated, how pure are the labels on which they agree, and what fraction of the data will survive the filter? Existing theory of disagreement-based learning [13]–[15] provides valuable sufficient conditions and bounds, but not a predictive law that can be computed from data and used to make design decisions.

This paper argues that such a law already exists, in an

unexpected place. On 7 October 2026 the Nobel Prize in Chemistry was awarded to Henri B. Kagan and Kenso Soai "for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis" [16]. Kagan showed in 1986 that the enantiomeric excess (ee) of a product need not be proportional to the ee of the chiral ligand of the catalyst [17]: when two ligands bind to one metal, mixed (heterochiral, "meso") complexes form and, if they are less reactive, the product is purer than the catalyst. His ML_2 model quantifies this amplification [18]–[20]. Soai later discovered a reaction whose product catalyzes its own formation, so that a small initial excess (2% in the original report) is amplified to 87% over successive reaction cycles [21], realizing the autocatalytic symmetry-breaking scheme proposed by Frank in 1953 [22], [23].

Our starting observation is that an agreement filter is, mathematically, an asymmetric catalyst. Each classifier "carries" a mixture of two enantiomers, its correct and its wrong predictions, whose excess is its label excess $\varepsilon = 2a - 1$ for accuracy a. When two classifiers are paired on an unlabeled point, a correct–correct or wrong–wrong pairing is a homochiral complex and a correct–wrong pairing is a heterochiral complex. Discarding disagreements is exactly the case of a catalytically inert meso complex, which is the textbook origin of a positive non-linear effect. Repeated self-training, in which the pseudo-labeled model generates the next round of pseudo-labels, is autocatalysis.

We make this analogy exact and use it. Our contributions are as follows.

1) **An exact correspondence theorem.** We prove that for two classifiers with arbitrary accuracies and arbitrary error correlation, the label excess of agreement-filtered pseudo-labels is identical to Kagan's ML_2 formula $\varepsilon_{\mathrm{out}} = \bar{\varepsilon}(1+\beta)/(1+g\beta)$, where β is the ratio of disagreeing to agreeing pairs and g is the probability with which disagreements are accepted (Theorem 2). For independent members the filter obeys a tanh law under which label log-odds are additive (Theorem 1).
2) **A predictive Chiral Amplification Law (CAL)** for committees of any size, any number of classes and correlated errors, which needs only three quantities that can be estimated on the labeled set: member accuracies, pairwise error correlation and wrong-label collision probability (Section IV-D). We characterize the optimal committee size (Proposition 2).
3) **A dynamical theory of autocatalytic self-training.** A mean-field model shows a pitchfork bifurcation at unit chiral susceptibility, the analogue of absolute asymmetric synthesis, and proves that amplification over rounds requires an effective committee size $M_{\mathrm{eff}} > 1$, i.e. diversity (Proposition 3).
4) **Identification and removal of calibration leakage.** We show that the out-of-fold accuracy estimates that self-training methods use to tune themselves are contaminated by pseudo-labels, and we introduce fold-specific pseudo-label chains that restore an honest estimate (Proposition 4). The honest estimate also enables a safe stopping rule.
5) **CAST, an algorithm, and an extensive empirical study.** Chiral Amplification Self-Training (CAST) selects its committee by maximizing the CAL-predicted noise-adjusted yield. We validate CAL on 20,400 real committees and compare CAST with two supervised references and four semi-supervised baselines, together with 7 ablations, on 34 public datasets using non-parametric statistical tests. All code, raw results and figure scripts are provided for reproducibility.

The remainder of the paper is organized as follows. Section II reviews related work. Section III formalizes the chemical analogy. Section IV develops the theory. Section V presents CAST. Sections VI and VII describe the experiments and results, Section VIII discusses limitations, and Section IX concludes. Proofs are collected in the Appendix.

II. RELATED WORK

A. SELF-TRAINING AND PSEUDO-LABELING

Self-training dates back to Scudder's adaptive pattern-recognition machines [3] and Yarowsky's word-sense disambiguation [4]. In deep learning it reappeared as pseudo-labeling [5] and is a core component of FixMatch [24], FlexMatch [25] and Noisy Student [26]. Because confirmation bias is the main failure mode [7], much work concerns which pseudo-labels to accept: fixed or class-adaptive confidence thresholds [24], [25], uncertainty estimates [27], [28] and, most recently, confidence–variance criteria [29]. Wei et al. [30] analyze self-training under expansion assumptions. These criteria are applied to a single model's scores; none predicts the precision of the accepted set before it is formed. Comprehensive surveys and benchmarks of self-labeling are given in [2], [6], and realistic evaluation practices are discussed in [31].

B. DISAGREEMENT-BASED METHODS

Co-training [8] trains two classifiers on two views and lets each label data for the other; its PAC analysis [13] and the expansion framework [14] explain why view independence helps, and Wang and Zhou [15] showed that a large diversity between learners is what matters, even with a single view. Tri-training [9] labels a point for one classifier when the other two agree and uses an error-rate condition to decide whether an update is beneficial. Democratic co-learning [10] combines different learning algorithms by weighted majority, and co-forest [11] extends the idea to random forests. Co-teaching [12] exchanges small-loss samples between two networks. Our theory explains quantitatively why diversity is the key quantity in all of these methods: it controls the fraction of heterochiral pairs, and therefore the size of the non-linear effect.

C. VOTING, DIVERSITY AND CORRELATED EXPERTS

The precision of a vote among independent experts is the subject of the Condorcet jury theorem; Ladha [32] studied

correlated votes, and Lam and Suen [33] analyzed majority voting for pattern recognition. Diversity measures for ensembles and their relation to ensemble accuracy were surveyed by Kuncheva and Whitaker [34], and the ambiguity decomposition of Krogh and Vedelsby [35] links ensemble error to member disagreement. Kittler et al. [36] derived combination rules under independence, and Dawid and Skene [37] estimated annotator error rates from agreement patterns. The independent-member precision formula in Theorem 1 is elementary in this literature; our contributions are its $\tanh / \operatorname{artanh}$ form, its exact generalization to correlated pairs, its identity with the ML_2 model, the multi-class collision correction, and the use of these results to drive pseudo-label selection.

D. LEARNING WITH NOISY LABELS AND SAFE SSL

Pseudo-labels are noisy labels whose noise is correlated with the learner's own errors. Angluin and Laird [38] showed that a sample with classification noise rate η is worth roughly $(1-2\eta)^2$ times as many clean examples; Natarajan et al. [39] developed loss corrections. We use the Angluin–Laird factor, which equals the squared label excess, as the objective for committee selection. Li and Zhou [40] studied when unlabeled data can be guaranteed not to hurt; our safe stopping rule pursues the same goal with an honest validation estimate.

E. LEAKAGE IN MODEL SELECTION

Information leakage in validation inflates performance estimates [41], [42]. Cross-fitting, in which nuisance estimates for one fold are computed without that fold, is standard in debiased estimation [43]. To our knowledge, the specific leak by which pseudo-labels transmit the labels of a validation fold back into its training folds has not been described for self-training; Section IV-G formalizes and removes it.

F. CHEMISTRY AND PHYSICS AS SOURCES OF LEARNING THEORY

Kagan's non-linear effects and their kinetic models are reviewed in [19], [20]; Soai's autocatalysis [21] and Frank's model [22] are reviewed in [23]. Statistical physics has repeatedly supplied learning theory, from Hopfield networks [44] to mean-field analyses; our bifurcation result is of Curie–Weiss type [45]. We are not aware of prior work connecting non-linear effects in asymmetric catalysis to pseudo-labeling.

III. THE CHEMICAL ANALOGY MADE PRECISE

A. NOTATION

Let $x \in \mathcal{X}$ be an input with true label $y \in \{1, \dots, C\}$. We observe a labeled set $\mathcal{L} = \{(x_i, y_i)\}_{i=1}^{n_L}$ and an unlabeled set $\mathcal{U} = \{x_j\}_{j=1}^{n_U}$ drawn from the same distribution, with $n_L \ll n_U$. A pool $\mathcal{H} = \{h_1, \dots, h_K\}$ of classifiers is available; a committee is a subset $S \subseteq \mathcal{H}$ of size $M = |S|$. Member h_i has accuracy $a_i = \Pr[h_i(x) = y]$ and error rate $b_i = 1 - a_i$. Table 1 lists the symbols used throughout.

TABLE 1. Main notation.

Symbol	Meaning
$\mathcal{L}$,$\mathcal{U}$	labeled and unlabeled sets (n_L, n_U points)
C	number of classes
$\mathcal{H}$, K	pool of base classifiers (catalysts) and its size
S, M	committee and its size
a_i, b_i	accuracy and error rate of member h_i
ε	label excess ("enantiomeric excess"), $\varepsilon = 2a-1$ for $C = 2$
λ	chiral logit, $\lambda = \ln(a/b) = 2\operatorname{artanh}\varepsilon$ for $C = 2$
ρ_{ij}	Pearson correlation between error indicators of h_i, h_j
κ_{ij}	collision: $\Pr[h_i(x) = h_j(x) \mid h_i, h_j$ both wrong$]$
$\pi(S)$, $Y(S)$	precision and yield (coverage) of the filter of S
β	ratio of disagreeing to agreeing pairs (meso/homo ratio)
g	acceptance rate of disagreements (meso reactivity)
M_{eff}	effective committee size, $M/(1+(M-1)\bar{\rho})$
$\Phi(S)$	effective yield, $Y(S)\,\varepsilon_{\text{out}}(S)^2$
$\mathcal{P}_t$	pseudo-label set after round t
s, χ	susceptibility to pseudo-labels; $\chi = sM_{\text{eff}}$

B. FROM ENANTIOMERS TO PREDICTIONS

In asymmetric synthesis a sample containing fractions p_R and p_S of two mirror-image molecules has enantiomeric excess $\mathrm{ee} = (p_R - p_S)/(p_R + p_S)$. We identify the correct label with the desired enantiomer R and any wrong label with S. For a binary classifier with accuracy a, the analogue of ee is the label excess

$$\varepsilon = a - b = 2a - 1 \in [-1, 1], \tag{1}$$

which is also the quantity $1 - 2\eta$ that governs learning from noisy labels at noise rate $\eta = b$ [38]. For $C > 2$ classes we use the excess over chance $\varepsilon = (Ca - 1)/(C - 1)$, which equals $2a - 1$ when $C = 2$.

Kagan's ML_2 model [17], [18] considers a metal M carrying two chiral ligands drawn from a non-enantiopure ligand pool. Three complexes form: $\mathrm{ML}_R\mathrm{L}_R$ and $\mathrm{ML}_S\mathrm{L}_S$ (homochiral), and the meso complex $\mathrm{ML}_R\mathrm{L}_S$ (heterochiral). The homochiral complexes produce opposite product enantiomers, while the meso complex produces a racemic product at relative rate g. With β the relative abundance of the meso complex, the product ee is

$$\mathrm{ee}_{\text{prod}} = \mathrm{ee}_0\,\mathrm{ee}_{\text{aux}}\,\frac{1+\beta}{1+g\beta}, \tag{2}$$

where ee_0 is the ee obtained with an enantiopure ligand. When $g < 1$ the product is purer than linear proportionality would suggest, a positive non-linear effect, (+)-NLE; when $g = 0$ the meso complex is a pure "reservoir" that removes the minor enantiomer [19], [20].

Fig. 1 and Table 2 show the correspondence. Two committee members evaluated on the same unlabeled point form a "complex". If both are right (RR) or both are wrong (SS), they agree, and the pseudo-label they emit carries their common chirality. If one is right and one is wrong (RS), they disagree. An agreement filter discards disagreements, which is precisely $g = 0$; a filter that resolves disagreements by a random or majority choice has $g > 0$.

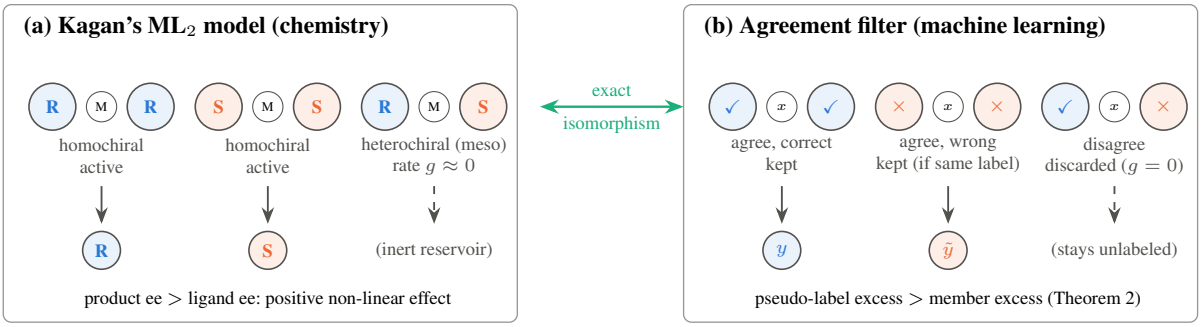


FIGURE 1. The chemical analogy. (a) In Kagan's ML_2 model, metal complexes carrying two ligands of equal chirality are catalytically active, while the mixed (meso) complex is inert, so the product is enantiomerically purer than the ligand pool. (b) In an agreement filter, two classifiers evaluated on the same unlabeled point x "complex" with it; pairs that agree emit a pseudo-label (correct y or wrong $\tilde{y}$), pairs that disagree are discarded. Theorem 2 shows that the two systems obey the same equation.

TABLE 2. Dictionary between asymmetric catalysis and agreement-filtered pseudo-labeling.

Asymmetric catalysis	Pseudo-labeling
Enantiomers R / S	correct / wrong prediction
Ligand ee, $\mathrm{ee}_{\mathrm{aux}}$	member label excess $\varepsilon = 2a - 1$
Product ee, $\mathrm{ee}_{\mathrm{prod}}$	pseudo-label excess $2\pi - 1$
Homochiral complex $\mathrm{ML}_R\mathrm{L}_R$ / $\mathrm{ML}_S\mathrm{L}_S$	two members agree (both right / both wrong)
Heterochiral (meso) complex	two members disagree
Meso/homo ratio β	disagreement/agreement ratio
Meso reactivity g	probability of accepting a disagreement
Product yield	filter yield (coverage) Y
Equilibrium constant of complexation	error correlation ρ (diversity)
Autocatalysis (Soai)	retraining on own pseudo-labels
Absolute asymmetric synthesis (Frank)	spontaneous labeling from $\varepsilon_0 = 0$

IV. THEORY

We first treat binary problems, where "wrong" is a single label, and then the multi-class case. Unless stated otherwise, probabilities are over a random unlabeled point; the classifiers are fixed. The unanimity filter of committee S accepts x if all members predict the same label and then emits that label. Its precision is $\pi(S) = \Pr[\hat{y} = y \mid \text{accept}]$ and its yield is $Y(S) = \Pr[\text{accept}]$.

A. THE TANH LAW

Theorem 1 (Tanh law)**.** *Let $C = 2$ and let the error events of the M members of S be mutually independent, with label excesses $\varepsilon_1, \ldots, \varepsilon_M \in (-1, 1)$. Then the unanimity filter has*

$$\varepsilon_{\mathrm{out}} = 2\pi(S) - 1 = \tanh\Big(\sum_{i\in S} \operatorname{artanh}\varepsilon_i\Big), \tag{3}$$

$$Y(S) = \prod_{i\in S} a_i + \prod_{i\in S} b_i. \tag{4}$$

For identical members, $\varepsilon_{\mathrm{out}} = \tanh(M \operatorname{artanh}\varepsilon)$ and $Y = 2^{1-M}(1-\varepsilon^2)^{M/2}\cosh(M \operatorname{artanh}\varepsilon)$.

In terms of the chiral logit $\lambda = \ln(a/b) = 2\operatorname{artanh}\varepsilon$, (3) states that chiral logits add: $\lambda_{\mathrm{out}} = \sum_i \lambda_i$. Since $\tanh$ is increasing and $\operatorname{artanh}\varepsilon_i > 0$ when $\varepsilon_i > 0$, the filter is always purer than each member, a positive non-linear effect, and it reaches enantiopurity only asymptotically. The price is yield: (4) decreases geometrically in M. Fig. 2(b) plots the law and Monte-Carlo verification.

B. EXACT CORRESPONDENCE WITH KAGAN'S MODEL

Real classifiers trained on the same data make correlated errors. The following result holds for any correlation.

Theorem 2 (Exact ML_2 correspondence)**.** *Let $C = 2$ and let two classifiers have accuracies a_1, a_2 and error indicators e_1, e_2 with Pearson correlation ρ. Suppose agreements are always accepted and disagreements are accepted with probability $g \in [0, 1]$ and then labeled by either member chosen uniformly at random. Then the accepted pseudo-labels have excess*

$$\varepsilon_{\mathrm{out}} = \bar{\varepsilon}\,\frac{1+\beta}{1+g\beta}, \qquad \bar{\varepsilon} = \frac{\varepsilon_1 + \varepsilon_2}{2}, \tag{5}$$

where $\beta = D/(1-D)$ is the ratio of disagreeing to agreeing pairs and

$$D = a_1 b_2 + a_2 b_1 - 2\rho\sqrt{a_1 b_1 a_2 b_2}. \tag{6}$$

This is Kagan's ML_2 formula (2) with $\mathrm{ee}_0 = 1$, $\mathrm{ee}_{\mathrm{aux}} = \bar{\varepsilon}$ and meso reactivity g.

The proof (Appendix) rests on a cancellation: the covariance term adds equally to correct–correct and wrong–wrong agreements, so the net excess of the agreeing pairs is always $\bar{\varepsilon}$, regardless of correlation; correlation acts only through β, exactly as complexation equilibria act in chemistry. Three consequences are immediate.

Corollary 1. *(i) The agreement filter ($g = 0$) amplifies by the factor $1 + \beta \geq 1$, with equality iff $D = 0$. (ii) For identical members ($a_1 = a_2$), $\varepsilon_{\mathrm{out}} = \varepsilon/[1 - (1-\rho)(1-\varepsilon^2)/2]$: amplification vanishes as $\rho \to 1$ and is maximal, $2/(1+\varepsilon^2)$, at $\rho = 0$ among non-negatively correlated pairs. (iii) Any rule that resolves disagreements by a coin flip ($g = 1$) has* no *non-linear effect: $\varepsilon_{\mathrm{out}} = \bar{\varepsilon}$.*

Part (ii) quantifies in closed form why diversity is the "key" to disagreement-based learning [15]: diversity is the

abundance of the meso complex. Part (iii) explains why accepting majority decisions in a two-member committee provides no purification. Fig. 2(a) shows the resulting Kagan plot and Fig. 3(a) the effect of g.

C. MULTI-CLASS PROBLEMS: COLLISIONS

With $C > 2$, two wrong members need not agree, which helps the filter. Let $\kappa_{ij} = \Pr[h_i(x) = h_j(x) \mid e_i = e_j = 1]$ be the collision probability of wrong predictions.

Proposition 1 (Multi-class filter)**.** *For two members, $\pi = P_{cc}/(P_{cc}+\kappa P_{ww})$ and $Y = P_{cc}+\kappa P_{ww}$, with $P_{cc} = a_1a_2 + \rho\sqrt{a_1b_1a_2b_2}$ and $P_{ww} = b_1b_2 + \rho\sqrt{a_1b_1a_2b_2}$. If M members err independently and uniformly over the $C-1$ wrong labels, $\kappa = 1/(C-1)$ and*

$$\ln\frac{\pi}{1-\pi} = \sum_{i\in S}\lambda_i + (M-1)\ln(C-1). \tag{7}$$

Each additional member thus contributes its chiral logit plus a "collision bonus" $\ln(C-1)$. In practice errors concentrate on confusable classes, so κ lies between $1/(C-1)$ and 1 and must be estimated.

D. THE CHIRAL AMPLIFICATION LAW FOR GENERAL COMMITTEES

For $M > 2$ correlated members, exact expressions require higher-order joint error probabilities that cannot be estimated reliably from a few labels. We therefore summarize correlation by the effective committee size, the design-effect formula of survey sampling [46]:

$$M_{\text{eff}}(S) = \frac{M}{1+(M-1)\bar{\rho}}, \qquad \bar{\rho} = \frac{2}{M(M-1)}\sum_{i<j\in S}\rho_{ij}, \tag{8}$$

which equals M for independent and 1 for identical members.

Definition 1 (Chiral Amplification Law, CAL)**.** *Let $\bar{\lambda}$ be the mean chiral logit of S, $\bar{a} = \sigma(\bar{\lambda})$ with σ the logistic function, and $\bar{\kappa}$ the mean pairwise collision probability ($\bar{\kappa} = 1$ if $C = 2$). The CAL predictions are*

$$\hat{\Lambda}(S) = M_{\text{eff}}\,\bar{\lambda} - (M_{\text{eff}}-1)\ln\bar{\kappa}, \qquad \hat{\pi}(S) = \sigma\big(\hat{\Lambda}(S)\big), \tag{9}$$

$$\hat{Y}(S) = \bar{a}^{\,M_{\text{eff}}} + (1-\bar{a})^{M_{\text{eff}}}\,\bar{\kappa}^{\,M_{\text{eff}}-1}. \tag{10}$$

CAL is exact in the limits that define it: it reduces to Theorem 1 for independent identical binary members, to (7) for symmetric multi-class errors, and to a single member when $\bar{\rho} = 1$. Between these limits it is an approximation whose accuracy we quantify in simulation (Fig. 4) and on real committees (Section VII-B).

E. EFFECTIVE YIELD AND THE OPTIMAL COMMITTEE SIZE

A larger committee produces purer but fewer pseudo-labels. Following Angluin and Laird [38], a label with noise rate η is worth about $(1-2\eta)^2 = \varepsilon^2$ clean labels, so the expected number of clean-equivalent labels per unlabeled point is the effective yield

$$\Phi(S) = Y(S)\,\varepsilon_{\text{out}}(S)^2. \tag{11}$$

This is the analogue of the product of chemical yield and squared optical purity.

Proposition 2 (Optimal committee size)**.** *For M identical, independent binary members with excess ε, $\Phi(M) = 2^{1-M}(1-\varepsilon^2)^{M/2}\cosh(M\theta)\tanh^2(M\theta)$ with $\theta = \operatorname{artanh}\varepsilon$. As $\varepsilon \to 0$, $\Phi(M)/\varepsilon^2 \to M^2 2^{1-M}$, which is maximized at $M^\star = 3$. Numerically, $M^\star \in \{2,3\}$ for all $\varepsilon \in (0,1)$, and $M^\star = 2$ for $\varepsilon \geq 0.351$.*

Proposition 2 predicts that small committees suffice under independence, and that weaker members call for slightly larger committees (Fig. 3(b)). Correlation lowers M_{eff} and therefore pushes the optimum toward larger nominal committees; CAST resolves the trade-off numerically by maximizing (11) over all committees.

F. AUTOCATALYSIS: A MEAN-FIELD MODEL OF REPEATED SELF-TRAINING

In self-training, the model retrained on pseudo-labels generates the next round of pseudo-labels, exactly as Soai's product catalyzes its own formation. We adopt the simplest model that captures this loop. Let ε_t be the label excess of the (homogeneous) members at round t and ε_0 their excess when trained on $\mathcal{L}$ alone. Retraining mixes the original knowledge with the pseudo-label signal of excess $\varepsilon_{\text{out}}(\varepsilon_t) = \tanh(M_{\text{eff}}\operatorname{artanh}\varepsilon_t)$:

$$\varepsilon_{t+1} = (1-s)\,\varepsilon_0 + s\,\tanh\big(M_{\text{eff}}\operatorname{artanh}\varepsilon_t\big), \tag{12}$$

where $s \in [0,1)$ is the susceptibility of the learner to pseudo-labels (larger for flexible learners and many pseudo-labels). We call $\chi = sM_{\text{eff}}$ the chiral susceptibility.

Proposition 3 (Amplification and bifurcation)**.** *Consider (12) with $M_{\text{eff}} \geq 1$ and $s \in (0,1)$.*

1) *If $\varepsilon_0 > 0$, the sequence ε_t increases monotonically to the unique fixed point $\varepsilon^\star \in [\varepsilon_0, 1)$, and $\varepsilon^\star > \varepsilon_0$ iff $M_{\text{eff}} > 1$. $\varepsilon^\star$ is increasing in M_{eff} and in s.*
2) *If $\varepsilon_0 = 0$ (no labeled information), $\varepsilon = 0$ is the only fixed point when $\chi \leq 1$; when $\chi > 1$ it becomes unstable and two symmetric stable fixed points $\pm\varepsilon^\star \neq 0$ appear: a pitchfork bifurcation at $\chi = 1$.*

Part 1 says that self-training amplifies the label excess only if the committee is effectively diverse, $M_{\text{eff}} > 1$; a committee of identical learners merely reproduces its own errors, which is confirmation bias [7] in quantitative form. Part 2 is the machine-learning analogue of Frank's absolute asymmetric synthesis and of Soai's experiments, in which the sign of the final excess is decided by chance: a sufficiently susceptible self-trainer will converge to a confident labeling even when it knows nothing, and that labeling may be the mirror image of the truth. Fig. 5 illustrates both regimes.

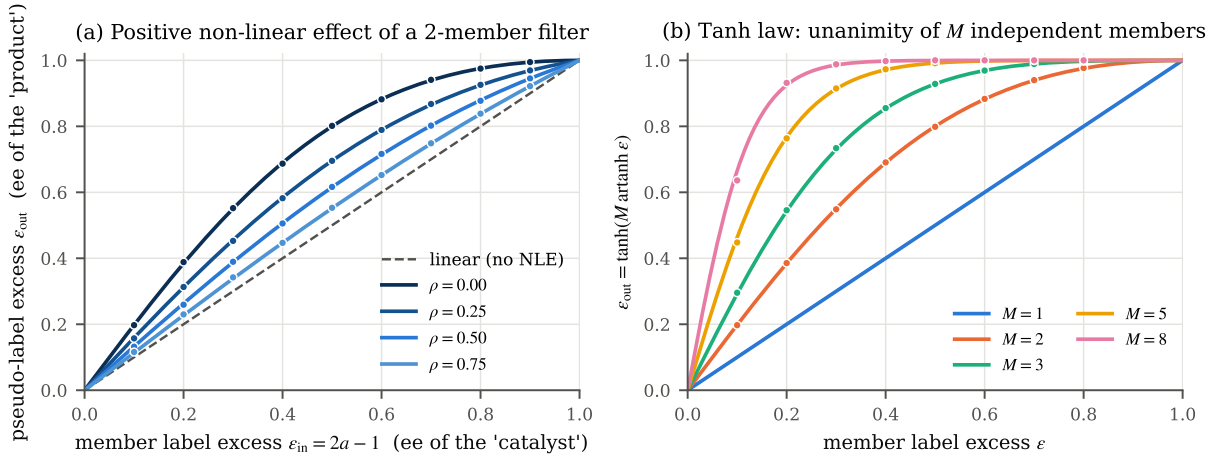


FIGURE 2. Non-linear effects of agreement filtering. (a) Kagan plot: label excess of two-member agreement-filtered pseudo-labels versus member label excess, for error correlations $\rho \in \{0, 0.25, 0.5, 0.75\}$. Curves: Theorem 2; dots: Monte-Carlo simulation ($N = 200{,}000$ points, max. deviation 0.0039). (b) Tanh law for unanimity of M independent members (Theorem 1); dots: simulation.

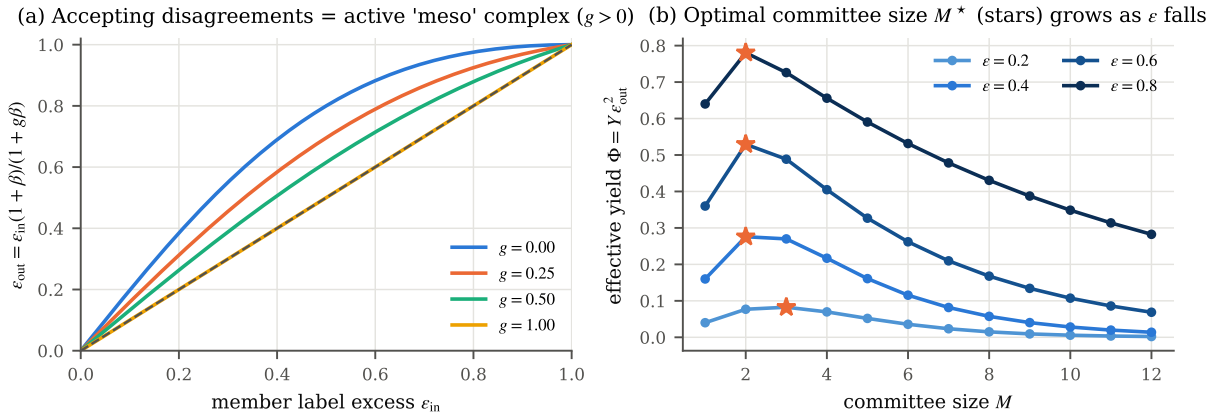


FIGURE 3. (a) Accepting a fraction g of disagreements corresponds to an active meso complex and erodes the non-linear effect; $g = 1$ is linear. (b) Effective yield $\Phi = Y\varepsilon_{out}^2$ versus committee size for independent members; stars mark the optimum $M^\star$ (Proposition 2).

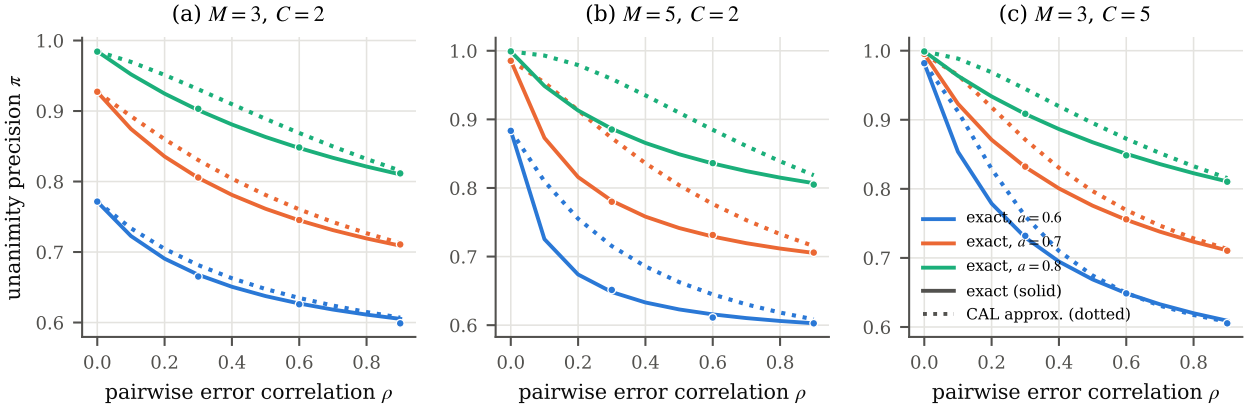


FIGURE 4. Accuracy of the CAL approximation (9) for $M > 2$ under a common-shock error model (with probability ρ all members share one error draw). Solid: exact precision; dotted: CAL; dots: simulation. The maximum absolute error is 0.030 for $M = 3$ binary, 0.097 for $M = 5$ binary and 0.058 for $M = 3$, $C = 5$; CAL is exact at $\rho = 0$ and $\rho = 1$ and least accurate at intermediate correlations.

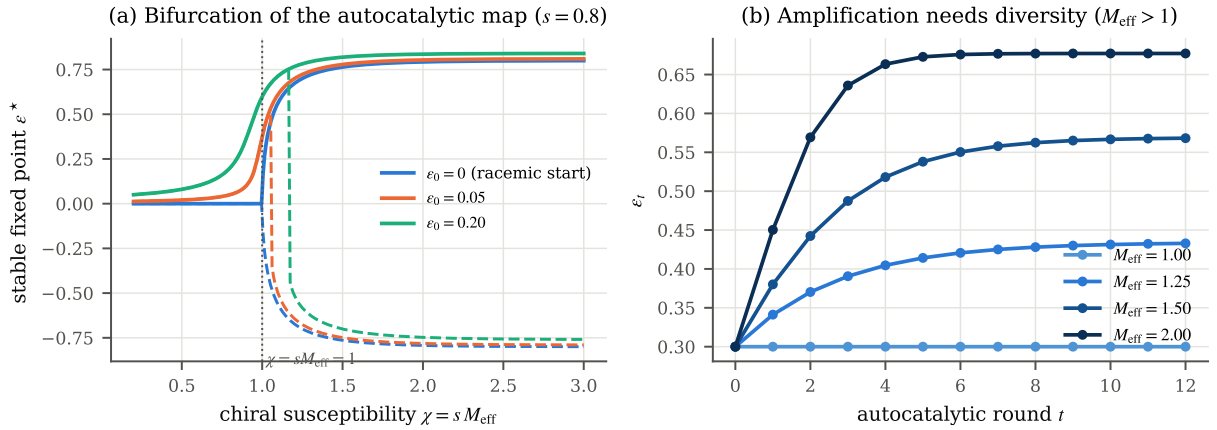


FIGURE 5. Autocatalytic mean-field map (12). (a) Stable fixed points versus chiral susceptibility $\chi = sM_{\text{eff}}$ (solid: positive branch, dashed: negative branch). Without labeled information ($\varepsilon_0 = 0$), a pitchfork bifurcation at $\chi = 1$ creates two mirror-image solutions (spontaneous symmetry breaking); any $\varepsilon_0 > 0$ selects the correct branch. (b) Trajectories from $\varepsilon_0 = 0.3$: amplification occurs only for $M_{\text{eff}} > 1$.

G. CALIBRATION LEAKAGE IN SELF-TRAINING

CAL needs a_i, ρ_{ij} and κ_{ij}, which we estimate from out-of-fold (OOF) predictions on $\mathcal{L}$. In round $t > 0$ the members are trained on $\mathcal{L}$ plus the current pseudo-labels $\mathcal{P}_{t-1}$. The natural estimator trains on $(\mathcal{L}\setminus V)\cup\mathcal{P}_{t-1}$ and evaluates on a fold $V \subset \mathcal{L}$. This estimator is leaky: $\mathcal{P}_{t-1}$ was produced by models that saw the labels of V, so the pseudo-labels of unlabeled points near V carry those labels back into training. The estimate then reflects memorization of V rather than generalization, and it inflates over rounds. We observed this directly during development, and it is quantified in Section VII-E: with leaky calibration, the OOF accuracy of the selected members exceeds their true accuracy on $\mathcal{U}$ by +0.081 on average after five rounds, and by up to +0.319 in individual runs, against +0.012 with the correction introduced below.

We remove the leak by fold-specific pseudo-label chains. For each fold f we maintain a separate pseudo-label set $\mathcal{P}_t^{(f)}$, generated by the same procedure as $\mathcal{P}_t$ but using only $\mathcal{L}\setminus V_f$.

Proposition 4 (Honest calibration)**.** *If fold f is predicted by models trained on $(\mathcal{L}\setminus V_f)\cup\mathcal{P}_{t-1}^{(f)}$, then, for fixed committees $S_1,\dots,S_{t-1}$, the training data of these models is a function of $(X_{\mathcal{L}}, y_{\mathcal{L}\setminus V_f}, X_{\mathcal{U}})$ only. Consequently, conditionally on the inputs and the committees, their OOF accuracy on V_f is an unbiased estimate of the accuracy that the same procedure achieves with $|\mathcal{L}\setminus V_f|$ labels, exactly as in ordinary cross-validation.*

The cost is one additional pseudo-labeling chain per fold. In return, every quantity used for decisions, CAL statistics, committee selection and the stopping rule, is computed without label leakage.

V. THE CAST ALGORITHM

CAST turns the theory of Section IV into a self-training algorithm. Fig. 6 shows its architecture, Fig. 7 its control flow, and Algorithm 1 the complete procedure. CAST has five components.

A. DIVERSE CATALYST POOL

The pool $\mathcal{H}$ contains $K = 7$ classifiers chosen to make different kinds of errors: logistic regression (LR), Gaussian naive Bayes (GNB), k-nearest neighbors (kNN, $k = 5$), a CART decision tree (DT), random forest (RF) [47], extremely randomized trees (ET) [48] and an RBF support-vector machine (SVM) [49]. Diversity across model families is what keeps the effective size M_{eff} above one (Proposition 3).

B. HONEST CHIRAL CALIBRATION

At the start of round t, CAST computes OOF predictions of every member on $\mathcal{L}$ with $k_f = 3$ stratified folds, using the fold-specific pseudo-label sets $\mathcal{P}_{t-1}^{(f)}$ of Proposition 4. From the $n_L \times K$ matrix of OOF predictions it estimates

$$\hat{a}_i = \frac{n_L - \#\text{err}_i + 1}{n_L + 2}, \qquad \hat{\rho}_{ij} = \text{corr}(e_i, e_j), \tag{13}$$
$$\hat{\kappa}_{ij} = \frac{\#\{e_i = e_j = 1,\ h_i = h_j\} + \alpha/(C-1)}{\#\{e_i = e_j = 1\} + \alpha},$$

with Laplace smoothing for accuracy and a prior of strength $\alpha = 2$ shrinking the collision probability toward the symmetric value $1/(C-1)$. The mean correlation in (8) is clipped to $[0, 1]$, a conservative choice that never credits a committee with more than M independent members.

C. COMMITTEE SELECTION BY EFFECTIVE YIELD

CAST evaluates the CAL predictions (9)–(10) for every committee of size $2 \le M \le K$, i.e. $2^K - K - 1 = 120$ candidates, and selects

$$S_t = \arg\max_S \hat{Y}(S)\ \max\{\hat{\varepsilon}_{\text{out}}(S), 0\}^2, \qquad \hat{\varepsilon}_{\text{out}} = \frac{C\hat{\pi} - 1}{C - 1}. \tag{14}$$

Because CAL needs only pairwise statistics, the search costs microseconds per candidate.

D. CROSS-FITTED UNANIMITY FILTERING AND AUTOCATALYTIC RELABELING

The members of S_t label the unlabeled set by cross-fitting: $\mathcal{U}$ is split into two halves, and each half is labeled by members trained on $\mathcal{L}$ plus the pseudo-labels of the other half only. Without this step, flexible learners (kNN, trees) would simply reproduce the pseudo-labels they were trained on, making the unanimity filter trivially satisfied on $\mathcal{P}_{t-1}$ and errors irreversible. Points on which all members of S_t agree form the candidate set. Within each class, candidates are ranked by the mean predicted probability of the committee and the top ones are kept, subject to a class-proportional quota equal to the class prior of $\mathcal{L}$ times a budget that grows linearly from $1/T$ of the candidate set in round 1 to all of it in round T.

The new pseudo-label set $\mathcal{P}_t$ is rebuilt from scratch every round, so a pseudo-label assigned early can be withdrawn or corrected later; this is the autocatalytic loop of Section IV, in which an improved model produces purer labels. Each pseudo-label receives the training weight $w_t = \max\{\hat{\varepsilon}_{\text{out}}(S_t), 0.05\}^2$, its Angluin–Laird value relative to a clean label. Labeled examples have weight one. The same step is carried out for every fold chain f using $\mathcal{L} \setminus V_f$.

E. SAFE ROUND SELECTION AND FINAL PREDICTOR

CAST comes in two variants that differ only in the final predictor, and each mirrors a supervised reference so that any difference is due to pseudo-labels alone. **CAST-1** uses the single pool member $h^\star$ with the best supervised OOF accuracy in round 0, exactly the rule of the supervised baseline Sup-Best. **CAST-V** uses the plurality vote of the whole pool, exactly the rule of the supervised baseline Sup-Ens. For CAST-1, CAST records the honest OOF accuracy of $h^\star$ after every round and selects

$$t^\star = \arg \max_{t \in \{0,\dots,T\}} \widehat{\text{acc}}_t(h^\star), \tag{15}$$

with ties broken toward fewer rounds; $t^\star = 0$ means that no pseudo-labels are used. $h^\star$ is then retrained on $\mathcal{L} \cup \mathcal{P}_{t^\star}$. CAST-V applies the same rule to the honest OOF accuracy of the pool vote (ties in the vote are broken by the round-0 ranking of the members) and retrains all members on $\mathcal{L} \cup \mathcal{P}_{t_V^\star}$. Fixing the predictor and selecting only the round limits the optimism of repeated comparisons, and the honesty of the estimates (Proposition 4) is what makes the rule meaningful. Both variants share the same pseudo-labeling process, so they can be computed in a single run.

F. COMPUTATIONAL COMPLEXITY

Let $c(n)$ be the cost of fitting one member on n points. Each round fits all K members k_f times for calibration and the $|S_t|$ committee members twice (one per half of $\mathcal{U}$) in each of the $k_f + 1$ chains, so the total cost is

$$O\big(T\,\big(k_f K + 2(k_f+1)|S_t|\big)\,c(n_L + n_U)\big)\,, \tag{16}$$

plus $O(T\,2^K K^2)$ for the committee search, which is negligible for $K = 7$. With $T = 5$, $k_f = 3$, $K = 7$ and the typical committee size of two to three members, CAST performs roughly 200–250 member fits per run, compared with about 22 for the supervised Sup-Best rule. Section VII-F reports measured run times.

Algorithm 1 Chiral Amplification Self-Training (CAST)

Require: labeled $\mathcal{L}$, unlabeled $\mathcal{U}$, pool $\mathcal{H}$ (K members), rounds T, folds k_f

Split $\mathcal{L}$ into stratified folds $V_1, \dots, V_{k_f}$
$\mathcal{P}_0 \leftarrow \emptyset$; $\mathcal{P}_0^{(f)} \leftarrow \emptyset$ for $f = 1..k_f$
for $t = 1, \dots, T+1$ **do**
  for $f = 1, \dots, k_f$ **do** ▷ honest calibration
    train every h_i on $(\mathcal{L} \setminus V_f) \cup \mathcal{P}_{t-1}^{(f)}$; predict V_f
  end for
  estimate $\hat{a}_i, \hat{\rho}_{ij}, \hat{\kappa}_{ij}$ by (13); record $\widehat{\text{acc}}_{t-1}$
  if $t = T+1$ **then break**
  end if
  $S_t \leftarrow \arg\max_S \hat{Y}(S)\hat{\varepsilon}_{out}(S)^2$ ▷ CAL, Eqs. (9)–(14)
  $\mathcal{P}_t \leftarrow \text{FILTER}(S_t, \mathcal{L}, \mathcal{P}_{t-1}, t)$
  for $f = 1, \dots, k_f$ **do** $\mathcal{P}_t^{(f)} \leftarrow \text{FILTER}(S_t, \mathcal{L} \setminus V_f, \mathcal{P}_{t-1}^{(f)}, t)$
  end for
end for
$h^\star \leftarrow \arg\max_i \widehat{\text{acc}}_0(h_i)$; $t^\star \leftarrow \arg\max_t \widehat{\text{acc}}_t(h^\star)$ ▷ CAST-1
$t_V^\star \leftarrow \arg\max_t \widehat{\text{acc}}_t(\text{pool vote})$ ▷ CAST-V
return $h^\star$ on $\mathcal{L} \cup \mathcal{P}_{t^\star}$ (CAST-1) or the pool vote trained on $\mathcal{L} \cup \mathcal{P}_{t_V^\star}$ (CAST-V)

function FILTER($S, \mathcal{A}, \mathcal{Q}, t$)
  split $\mathcal{U}$ into halves $\mathcal{U}_1, \mathcal{U}_2$
  label $\mathcal{U}_1$ ($\mathcal{U}_2$) with members of S trained on $\mathcal{A} \cup \mathcal{Q}|_{\mathcal{U}_2}$ ($\mathcal{A} \cup \mathcal{Q}|_{\mathcal{U}_1}$)
  $\mathcal{G} \leftarrow \{x \in \mathcal{U} :$ all members of S agree$\}$
  keep the $\lceil (t/T)\,|\mathcal{G}|\,p_c \rceil$ most confident points of each class c
  return kept points with their labels and weight $\max\{\hat{\varepsilon}_{out}(S), 0.05\}^2$
end function

VI. EXPERIMENTAL SETUP

A. DATASETS

We use 34 public classification datasets: 33 tabular datasets from the UCI repository [50], obtained from a public ARFF mirror,[1] and the Fashion-MNIST image dataset [51]. The tabular datasets were selected a priori by a single criterion, at least 300 instances, without looking at any result. Table 3 lists their characteristics; they span 2 to 26 classes, 4 to 286 features, balanced and highly imbalanced class distributions, and numeric, nominal and mixed attributes.

Preprocessing is identical for all datasets. Nominal attributes are one-hot encoded (attributes with more than 50 levels, which are identifiers, are dropped), missing values are imputed with the median or the mode, classes with fewer than 20 instances are removed so that every class can be represented in every labeled fold, and all features are standardized. Datasets with more than 2,000 instances are subsampled to 2,000 instances by stratified sampling with a fixed seed. For Fashion-MNIST we draw 2,000 images from the training split and project the pixels onto 50 principal components; the projection uses no labels.

[1] https://github.com/renatopp/arff-datasets

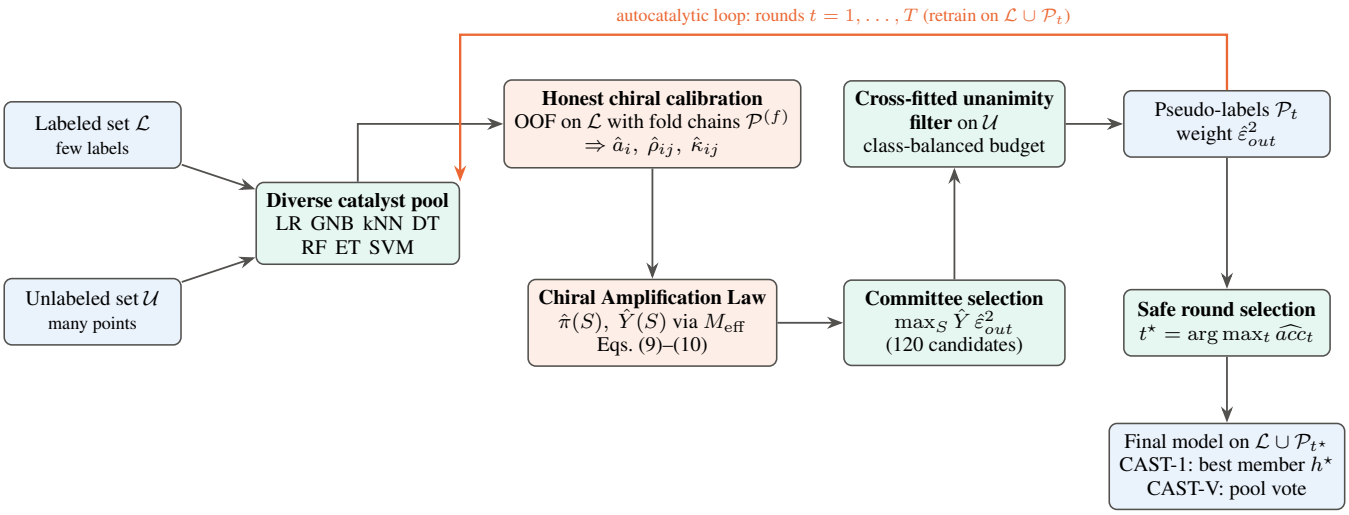


FIGURE 6. Architecture of Chiral Amplification Self-Training (CAST). A diverse pool of classifiers (the "catalysts") is calibrated without label leakage, the Chiral Amplification Law predicts the purity and yield of every committee, the committee with the largest effective yield labels the unlabeled data by cross-fitted unanimity, and the pseudo-labels are fed back to the pool (autocatalysis). A safe rule decides how many rounds to keep.

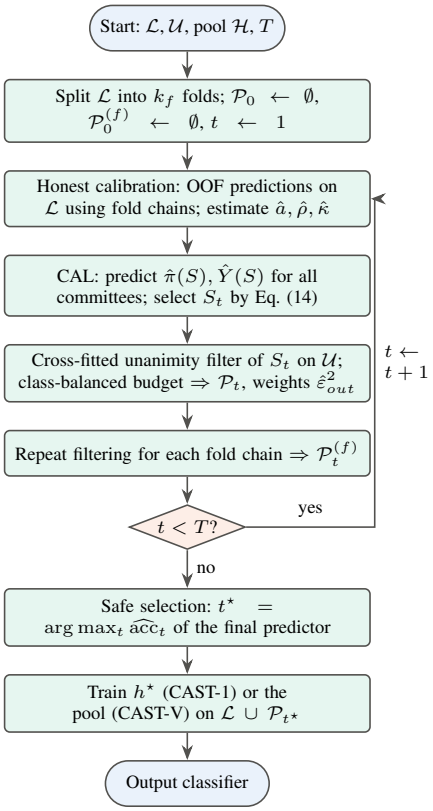


FIGURE 7. Flowchart of one CAST run.

B. PROTOCOL

For every dataset and random seed $r \in \{0, \ldots, 4\}$ we draw a stratified 75/25 train/test split. Within the training part, a stratified fraction $\gamma \in \{10\%, 20\%\}$ of each class forms $\mathcal{L}$ (at least three examples per class) and the remainder forms $\mathcal{U}$ with its labels hidden. All methods receive exactly the same $\mathcal{L}, \mathcal{U}$ and test set. The test set is never used for any decision; it serves only for the reported accuracy and macro-F1. In total the main benchmark comprises 3,060 method runs.

TABLE 3. Datasets after preprocessing: instances n, features d, classes C, majority-class share and imbalance ratio (IR, largest/smallest class).

Dataset	n	d	C	Maj.	IR	Dataset	n	d	C	Maj.	IR
anneal	890	33	4	0.77	17.1	letter	2000	16	26	0.04	1.1
balance.scale	625	4	3	0.46	5.9	mushroom	2000	108	2	0.52	1.1
breast.w	699	9	2	0.66	1.9	nursery	2000	26	4	0.33	13.1
car	1728	21	4	0.70	18.6	optdigits	2000	61	10	0.10	1.0
cmc	1473	21	3	0.43	1.9	page.blocks	2000	10	5	0.90	179.6
colic	368	60	2	0.63	1.7	pendigits	2000	16	10	0.10	1.1
column3C	310	6	3	0.48	2.5	segment	2000	18	7	0.14	1.0
credit.a	690	42	2	0.56	1.2	sick	2000	31	2	0.94	15.4
credit.g	1000	59	2	0.70	2.3	soybean	630	82	15	0.15	4.6
cylinder.bands	540	83	2	0.58	1.4	spambase	2000	57	2	0.61	1.5
dermatology	366	129	6	0.31	5.6	splice	2000	286	3	0.52	2.2
diabetes	768	8	2	0.65	1.9	tic-tac-toe	958	27	2	0.65	1.9
ecoli	327	6	5	0.44	7.2	vehicle	846	18	4	0.26	1.1
heart.c	303	22	2	0.54	1.2	vote	435	16	2	0.61	1.6
hypothyroid	2000	30	3	0.92	36.9	vowel	990	27	11	0.09	1.0
ionosphere	351	33	2	0.64	1.8	waveform5000	2000	40	3	0.34	1.0
kr.vs.kp	2000	38	2	0.52	1.1	fashion-mnist	2000	50	10	0.11	1.1

C. METHODS COMPARED

We compare CAST with two supervised references and four semi-supervised methods that represent the main families of self-labeling:

- **Sup-Best**: the pool member with the best OOF accuracy on $\mathcal{L}$, trained on $\mathcal{L}$; CAST uses the same member-selection rule.
- **Sup-Ens**: plurality vote of the whole pool trained on $\mathcal{L}$.
- **ST-RF**: classic confidence-threshold self-training [3], [4] with a 50-tree random forest, threshold 0.75 and up to 10 iterations (scikit-learn implementation).
- **Tri-Train**: tri-training [9] with 50-tree random forests as base learners, implemented following the original pseudo-code including its error-rate update conditions.

- **LabelSpread**: graph-based label spreading [52] with a 7-nearest-neighbor kernel and $\alpha = 0.2$.
- **Maj-ST**: democratic-style self-training [10], which accepts a pseudo-label when a strict majority of the whole pool agrees. To isolate the effect of the filter, Maj-ST shares all other machinery of CAST (honest calibration, cross-fitting, relabeling, budget, safe round selection, final predictors); in the language of Theorem 2 it is the $g > 0$ regime.

Both committee methods are evaluated with the two final predictors of Section V: Maj-ST-1 and CAST-1 (single member, compare with Sup-Best) and Maj-ST-V and CAST-V (pool vote, compare with Sup-Ens), for a total of 9 methods. Deep SSL methods such as FixMatch [24] rely on data augmentation of images and are not applicable to tabular data; we therefore compare with self-labeling methods of the kind benchmarked in [6].

D. ABLATIONS

To test each component, we run 7 variants of CAST on the 10% setting with seeds 0–2: **CAST-All** (no committee selection; unanimity of the whole pool), **CAST-Indep** (selection with the independence law, $\rho = 0$, $\kappa = 1/(C-1)$), **CAST-Emp** (selection with raw OOF counts instead of CAL), **CAST-1R** (a single round, no autocatalysis), **CAST-Homog** (a homogeneous pool of seven bootstrap CART trees), **CAST-Leaky** (naive, leaky calibration), and **CAST-Unsafe** (always uses the last round, $t^\star = T$).

E. IMPLEMENTATION AND HYPER-PARAMETERS

All methods are implemented in Python with scikit-learn [53]. CAST uses $T = 5$ rounds, $k_f = 3$ calibration folds, 30 trees for RF and ET, and default scikit-learn settings otherwise. These values, and all design decisions of CAST, were fixed during development on pilot seeds (100 and 101) on a subset of 12 datasets, which are disjoint from the evaluation seeds 0–4. No hyper-parameter was tuned on the evaluation seeds or on any test set. For transparency we note one exception to the order of work: the pool-vote predictor (the -V variants) was added after the first evaluation runs had shown that the supervised ensemble Sup-Ens was the strongest reference; it introduces no new hyper-parameter, was checked on the pilot seeds before being run on the evaluation seeds, and we report the single-member variants alongside it. Experiments ran on a 2-core CPU.

F. STATISTICAL ANALYSIS

Following Demšar [54], we rank the methods on each dataset by mean test accuracy over seeds, compare average ranks with the Friedman test [55] and visualize them in a critical-difference diagram based on the Nemenyi test. Pairwise comparisons of CAST against each other method use the Wilcoxon signed-rank test [56] over datasets with Holm's correction [57]. We also report win/tie/loss counts.

TABLE 4. Prediction of unanimity-filter precision π and yield Y for 20,400 real committees from labeled data only (10% labels). MAE: mean absolute error; Bias: mean signed error; r_s: Spearman correlation; Regret: observed effective yield of the selected committee relative to the best committee in hindsight (1 is optimal).

	Precision π			Yield Y		Selection
Predictor	MAE	Bias	r_s	MAE	r_s	Regret
CAL (ours)	0.069	-0.043	0.820	0.095	0.853	0.914
Independence law	0.084	+0.082	0.764	0.237	0.820	0.914
Empirical OOF counts	0.051	-0.032	0.856	0.066	0.913	0.919
Whole pool (no selection)	–	–	–	–	–	0.641

VII. RESULTS

A. VERIFICATION OF THE THEORY BY SIMULATION

Before turning to real data, we verified the closed-form results by Monte-Carlo simulation with $N = 200{,}000$ points per configuration. The two-member formula of Theorem 2 matched simulation to within 0.0039 for all tested correlations (Fig. 2(a)), and the tanh law matched to within 0.030 for up to $M = 8$ members (Fig. 2(b)), the residual being sampling noise at low yield. For $M > 2$ correlated members, the CAL approximation (9) deviates from the exact common-shock precision by at most 0.030 for $M = 3$ and 0.097 for $M = 5$ binary members, and by at most 0.058 for $M = 3$ members with $C = 5$ classes (Fig. 4).

B. DOES CAL PREDICT REAL COMMITTEES?

For each of the 34 datasets and five seeds (10% labels), we trained the seven pool members on $\mathcal{L}$ and formed all 120 committees of size two to seven, giving 20,400 committee instances. For each one we compared the precision and yield of its unanimity filter on $\mathcal{U}$ (using the hidden labels) with three predictions computed from $\mathcal{L}$ alone: CAL, the independence law, and the raw OOF frequency of unanimous-and-correct predictions. Table 4 and Fig. 8 report the results.

Three findings stand out. First, the correlation and collision corrections of CAL matter: compared with the independence law, CAL reduces the precision MAE from 0.084 to 0.069 and the yield MAE from 0.237 to 0.095. The independence law is systematically over-confident (bias +0.082), exactly as Corollary 1 predicts for positively correlated members, whereas CAL is slightly conservative (bias -0.043). Second, CAL's structural approximation is accurate: when the three pairwise statistics are measured on $\mathcal{U}$ instead of being estimated from $\mathcal{L}$, CAL predicts precision with MAE 0.029 and Spearman correlation 0.96. Most of its error on real data therefore comes from estimating $\hat{a}$, $\hat{\rho}$ and $\hat{\kappa}$ from a few dozen labels, not from the law itself. For two-member binary committees, where Theorem 2 is exact, the exact formula with estimated statistics attains MAE 0.045 over 1470 instances.

Third, the raw empirical OOF frequency is a strong competitor. Over all committees it predicts precision more accurately (MAE 0.051) than CAL, because it uses the full joint behavior of the committee rather than pairwise summaries. Table 5 shows, however, that the advantage depends on

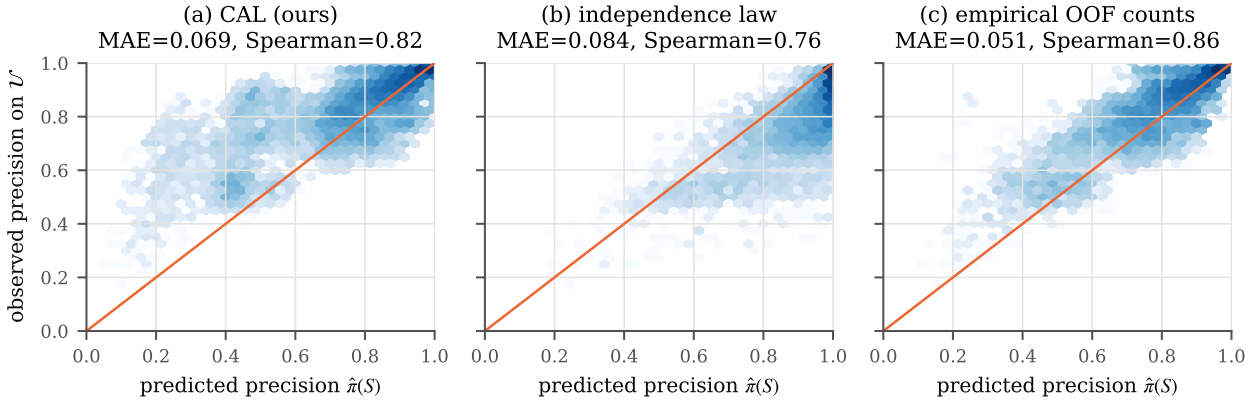


FIGURE 8. Predicted versus observed precision of the unanimity filter for 20,400 real committees (color: log count). Predictions use only the labeled set (10% labels); observations use the hidden labels of $\mathcal{U}$.

TABLE 5. Precision MAE by size of the labeled set (terciles over dataset/seed pairs). Oracle: CAL with statistics measured on $\mathcal{U}$.

Labeled-set size	Committees	CAL	Indep.	Emp.	Oracle
Small ($22 \leq n_L \leq 52$)	6600	0.061	0.088	0.065	0.012
Medium ($58 \leq n_L \leq 149$)	6600	0.099	0.123	0.065	0.049
Large ($150 \leq n_L \leq 156$)	7200	0.048	0.044	0.027	0.025

TABLE 6. Precision MAE by committee size M (all datasets and seeds, 10% labels).

M	Committees	Mean π	CAL	Indep.	Emp.
2	3570	0.846	0.063	0.073	0.052
3	5950	0.874	0.067	0.086	0.051
4	5950	0.890	0.070	0.087	0.051
5	3570	0.900	0.072	0.086	0.052
6	1190	0.908	0.074	0.083	0.053
7	170	0.914	0.076	0.080	0.055

the amount of labeled data: with the smallest labeled sets ($n_L \leq 52$), where unanimous OOF events are rare, CAL is the more accurate predictor (0.061 versus 0.065). For committee selection, which is what CAST needs, all three predictors are nearly equivalent and recover about 0.914–0.919 of the best achievable effective yield, whereas using the whole pool achieves only 0.641. In short, CAL is accurate where it matters, is interpretable, and has the extra advantage of being an explicit function of accuracy, diversity and collisions, which is what makes the design insights of Section IV possible.

Table 6 confirms two predictions of the theory on real classifiers: the observed precision of unanimity rises monotonically with committee size, and the error of the independence law does not shrink with M because correlated members contribute less than one "independent vote" each.

C. NON-LINEAR EFFECTS IN REAL COMMITTEES

Fig. 9(a) plots, for every CAST run with 10% labels, the label excess of the first-round pseudo-labels against the mean label excess of the selected committee members, both measured on $\mathcal{U}$. In 100.0% of the runs the point lies above the diagonal, i.e. the agreement filter produced a positive non-linear effect, as Theorem 2 predicts whenever members are not perfectly correlated. On average the member excess was 0.695 and the pseudo-label excess 0.788, with a mean estimated pairwise error correlation of 0.47 and effective committee size of 1.44. CAST selected two-member committees in 65.6% of the rounds, three-member committees in 24.8%, and larger ones in 9.5% (mean size 2.46), in line with Proposition 2. The most frequently selected members were LR (53%), ET (44%), RF (39%).

Fig. 9(b) follows CAST over its rounds. The precision of the filter stays high (0.865 in round 1, 0.859 in round 5) while the budget grows to include all unanimous points, and the CAL prediction tracks the observed precision closely throughout (mean predicted minus observed precision +0.000). The accuracy of the committee members on $\mathcal{U}$ moves from 0.803 to 0.812, a modest autocatalytic gain that is consistent with the mean-field model when the susceptibility s is small, as expected for learners that also see clean labels.

D. CLASSIFICATION ACCURACY

Table 7 summarizes the benchmark over all 34 datasets and five seeds, Tables 8 and 9 give the per-dataset results for 10% and 20% labels, and Fig. 10 shows the critical-difference diagrams. We report two versions of each committee method: with the single-member predictor (suffix -1, same predictor as Sup-Best) and with the pool-vote predictor (suffix -V, same predictor as Sup-Ens).

The Friedman test rejects the hypothesis that all methods perform equally ($p = 8.0 \times 10^{-17}$ at 10% and $p = 9.3 \times 10^{-15}$ at 20% labels). At both label ratios, CAST-V attains the highest mean accuracy (82.17% and 85.30%), the highest macro-F1 (76.54% and 80.58%) and the best average rank (3.07 and 3.16). It is significantly more accurate than the classical self-labeling methods after Holm correction: ST-RF ($p_{\mathrm{Holm}} < 0.001$), tri-training (0.002) and label spreading

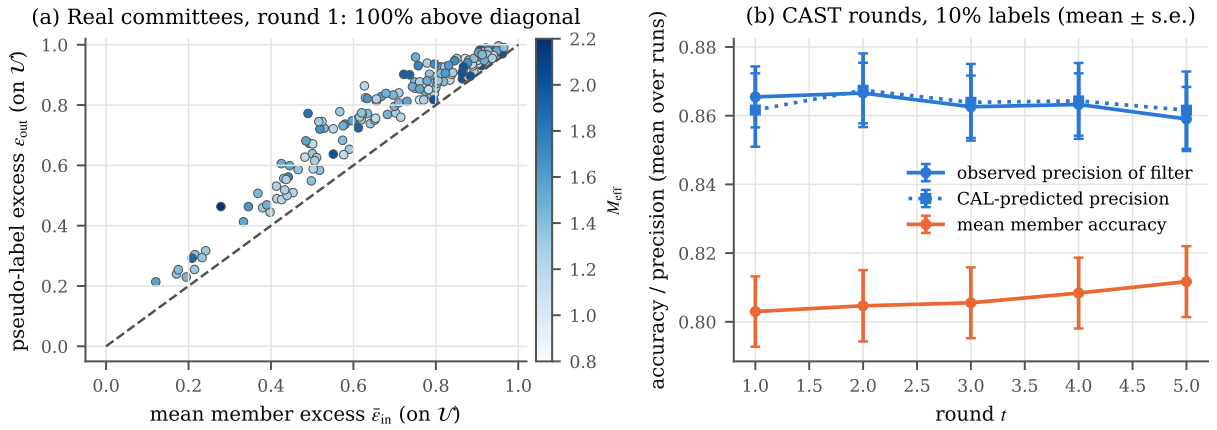


FIGURE 9. (a) Positive non-linear effect on real data: label excess of first-round pseudo-labels versus mean member excess on $\mathcal{U}$, colored by the estimated effective committee size. (b) Evolution over autocatalytic rounds of the observed precision of the unanimity filter, its CAL prediction, and the accuracy of the committee members on $\mathcal{U}$ (mean ± standard error over runs).

TABLE 7. Summary over 34 datasets (mean of five seeds per dataset). W/T/L: datasets on which CAST-V wins/ties/loses against the method; p: Wilcoxon signed-rank test; p_{Holm}: Holm-corrected.

Method	Acc. (%)	F1 (%)	Rank	W/T/L	p	p_{Holm}
(a) 10% labeled data						
Sup-Best	80.83	75.02	5.46	24/1/9	0.005	0.024
Sup-Ens	81.65	75.11	3.78	23/0/11	0.032	0.128
ST-RF	77.88	68.02	6.63	28/0/6	<0.001	<0.001
Tri-Train	80.39	71.50	5.81	28/0/6	<0.001	0.002
LabelSpread	72.76	65.29	7.94	33/0/1	<0.001	<0.001
Maj-ST-1	81.51	76.15	4.37	21/0/13	0.032	0.128
Maj-ST-V	81.57	75.96	3.41	23/1/10	0.053	0.128
CAST-1	81.40	76.09	4.53	21/1/12	0.043	0.128
CAST-V	**82.17**	**76.54**	**3.07**	–	–	–
(b) 20% labeled data						
Sup-Best	84.64	79.85	5.57	25/0/9	0.012	0.060
Sup-Ens	84.82	79.54	3.76	19/3/12	0.067	0.164
ST-RF	82.67	75.56	6.47	26/0/8	<0.001	<0.001
Tri-Train	84.39	78.46	5.03	25/0/9	0.002	0.011
LabelSpread	76.81	69.60	7.97	32/0/2	<0.001	<0.001
Maj-ST-1	84.89	80.43	4.84	24/1/9	0.018	0.072
Maj-ST-V	84.81	79.96	3.63	22/1/11	0.055	0.164
CAST-1	85.16	80.54	4.56	23/0/11	0.087	0.164
CAST-V	**85.30**	**80.58**	**3.16**	–	–	–

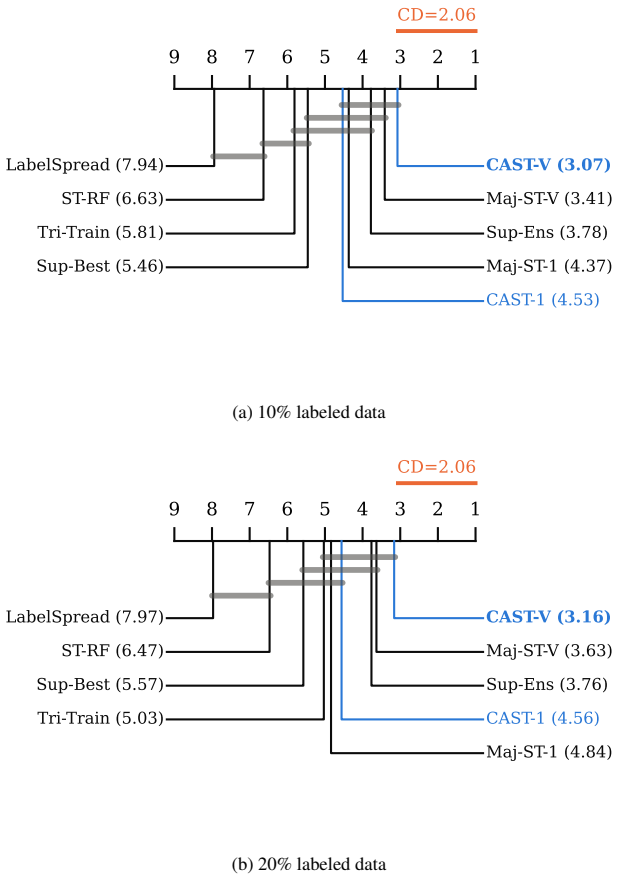


FIGURE 10. Critical-difference diagrams (Nemenyi test, $\alpha = 0.05$). Lower rank is better; methods joined by a gray bar are not significantly different. CD $= 2.06$ (a) and 2.06 (b).

(<0.001) at 10% labels, with the same conclusion at 20%. Its advantage over the supervised single-member reference Sup-Best is significant at 10% ($p_{\text{Holm}} = 0.024$) but not after correction at 20% (0.060). Against the strongest references, the supervised ensemble Sup-Ens and the democratic-style Maj-ST-V, CAST-V wins on most datasets (Table 7) and the uncorrected Wilcoxon p-values are 0.032 and 0.053 at 10%, but the differences are not significant after Holm correction, and the critical-difference diagrams (Fig. 10) place these methods in the same group. We therefore read the accuracy results as consistent but modest improvements.

Classical self-labeling often hurts. With the predictor held fixed (Table 10), confidence-threshold self-training (ST-RF) loses -2.96 points on average relative to Sup-Best at 10% labels and is worse on 23 of the 34 datasets; it loses more than two points in 50.0% of the individual runs, and label spreading in 66.5%. This is the confirmation bias that motivates agreement filtering. CAST-V, in contrast, loses more than two points against Sup-Ens in only 11.2% of the runs at 10% labels and 6.5% at 20%. One reason is the safe

TABLE 8. Test accuracy (%, mean ± standard deviation over five seeds) with 10% labeled data. Best per dataset in bold.

Dataset	Sup-Best	Sup-Ens	ST-RF	Tri-Train	LabelSpread	Maj-ST-1	Maj-ST-V	CAST-1	CAST-V
anneal	79.9±2.4	81.3±3.7	**82.2**±3.0	79.8±2.0	61.1±6.1	80.6±1.9	81.4±2.0	79.4±2.9	81.6±1.5
balance.scale	84.7±2.0	81.9±2.1	70.6±3.1	74.6±2.0	78.1±2.9	81.5±6.7	79.1±4.2	**85.0**±1.5	81.8±2.5
breast.w	93.7±2.8	96.2±1.1	93.7±2.6	94.4±0.4	95.0±1.8	93.7±2.8	96.1±1.2	93.8±2.9	**96.6**±0.8
car	83.4±1.8	**84.5**±1.7	76.6±2.2	81.9±1.2	74.1±0.9	83.5±1.8	83.8±2.7	83.5±1.8	84.1±3.0
cmc	43.3±1.7	44.4±2.2	45.0±1.7	**46.0**±2.9	39.0±3.4	43.3±1.7	43.6±2.2	43.6±1.2	43.4±1.7
colic	70.7±8.4	74.3±6.8	69.3±5.1	72.4±3.1	60.2±11.3	74.3±7.3	74.1±5.6	69.3±8.6	**75.0**±4.8
column3C	79.7±3.1	79.5±0.8	80.5±2.2	**81.0**±4.9	73.6±1.7	80.5±2.9	80.0±2.6	79.7±3.1	79.2±2.7
credit.a	78.4±6.1	**86.1**±1.0	82.7±2.9	84.2±3.0	62.3±8.4	79.1±6.5	85.9±0.9	78.0±6.2	85.9±1.0
credit.g	69.4±1.6	**70.8**±1.6	69.9±0.4	69.8±0.3	68.0±1.7	69.4±1.6	70.6±2.0	69.3±1.6	70.3±2.4
cylinder.bands	65.3±4.6	66.4±2.7	61.3±2.5	63.4±3.7	61.0±6.2	**67.4**±4.8	64.1±3.7	67.0±4.6	67.1±5.3
dermatology	90.4±5.1	95.9±2.1	89.6±2.2	95.0±2.6	81.1±6.2	94.3±1.9	**98.0**±1.3	92.2±6.1	97.4±1.5
diabetes	70.5±1.9	70.5±2.1	68.5±2.6	69.4±2.5	68.1±3.0	70.3±1.8	70.5±2.2	**70.8**±2.0	70.0±2.2
ecoli	86.6±3.6	84.9±4.0	81.2±3.1	82.2±2.3	85.6±3.6	86.8±3.2	**87.1**±2.3	**87.1**±3.6	86.3±2.4
heart.c	79.5±4.0	79.7±2.4	72.6±5.7	76.1±6.9	73.9±8.0	80.5±4.1	**81.3**±2.8	80.0±5.1	79.5±3.5
hypothyroid	96.9±1.0	94.3±0.4	93.8±0.5	94.0±0.5	84.9±2.7	96.9±1.0	94.3±0.4	**97.3**±1.5	95.0±1.5
ionosphere	85.7±3.6	86.6±2.0	**88.6**±4.1	83.6±5.6	70.0±8.1	84.8±5.1	86.6±4.3	85.0±4.7	87.0±4.1
kr.vs.kp	92.3±1.9	92.3±1.6	92.2±2.0	91.8±1.6	72.9±1.9	**92.8**±1.0	92.4±1.8	92.1±1.3	92.6±1.5
letter	54.4±1.9	56.9±1.0	52.1±1.3	56.8±1.1	52.4±1.7	55.0±2.5	54.5±1.9	57.1±2.6	**59.7**±1.3
mushroom	98.2±0.9	98.5±0.6	98.5±0.6	98.0±1.2	98.2±0.6	98.3±0.9	98.5±0.6	98.2±0.8	**98.6**±0.5
nursery	85.7±2.1	86.8±0.9	81.8±4.2	85.2±0.9	65.9±1.4	85.7±1.9	86.2±1.6	**87.0**±1.1	86.8±0.9
optdigits	91.8±2.1	92.7±1.9	85.8±2.2	91.6±1.5	94.1±0.9	92.8±1.1	94.3±1.0	93.8±1.2	**95.2**±0.7
page.blocks	95.0±0.9	95.6±0.9	95.4±0.6	95.5±0.5	94.8±0.8	94.6±1.1	95.7±1.4	94.8±1.2	**95.9**±0.8
pendigits	90.8±1.8	90.6±1.6	85.4±2.1	89.4±2.0	**93.6**±1.7	91.6±0.9	90.8±1.5	90.9±2.0	90.9±1.7
segment	92.0±0.6	92.2±1.1	91.5±1.1	91.7±0.6	86.0±3.1	92.5±1.2	**93.0**±0.9	92.7±0.7	92.3±1.0
sick	95.6±1.2	95.6±1.2	94.0±0.2	94.0±0.1	94.0±1.3	95.7±1.1	94.7±1.4	**95.8**±1.1	95.0±1.8
soybean	82.7±2.1	83.2±2.5	79.9±3.9	80.4±1.1	83.0±3.3	83.8±1.9	84.2±2.2	83.5±2.5	**85.4**±2.3
spambase	89.2±0.9	91.2±0.7	90.0±1.2	91.3±1.0	79.1±2.7	90.4±1.6	91.6±0.5	90.1±1.4	**91.7**±0.6
splice	87.8±2.6	91.7±1.3	54.4±0.9	81.8±5.0	57.0±1.4	92.5±1.4	94.1±0.8	91.6±1.6	**94.4**±1.1
tic-tac-toe	74.8±3.9	**77.8**±1.6	71.8±0.8	74.8±2.1	63.6±3.0	74.4±3.8	74.8±3.9	73.8±3.5	74.2±3.2
vehicle	67.5±4.9	**67.6**±3.6	59.7±4.0	67.3±4.1	60.8±3.2	67.1±4.8	64.3±4.7	67.0±4.6	66.6±2.8
vote	93.4±3.2	93.6±2.5	**95.0**±2.8	92.7±2.8	83.7±6.0	93.4±3.2	93.6±2.5	93.4±3.2	94.1±2.4
vowel	**52.4**±4.1	39.8±3.7	51.0±2.9	52.3±4.2	24.7±1.3	51.0±2.6	39.8±3.7	51.9±3.6	45.5±3.0
waveform5000	79.8±1.3	82.0±1.0	77.4±1.5	83.0±1.1	69.2±0.7	81.4±3.0	**83.3**±1.6	81.3±2.1	82.7±1.1
fashion-mnist	67.0±1.5	70.5±2.2	65.5±1.3	68.0±1.6	64.7±1.6	71.2±2.9	70.9±1.1	71.4±3.1	**71.8**±1.9
Avg. acc.	80.83	81.65	77.88	80.39	72.76	81.51	81.57	81.40	82.17
Avg. rank	5.46	3.78	6.63	5.81	7.94	4.37	3.41	4.53	3.07
#Best	1	5	3	2	1	2	5	6	10

round selection: CAST-1 keeps the purely supervised model ($t^\star = 0$) in 45.3% of its runs, i.e. it declines to use pseudo-labels whenever honest calibration does not show a benefit.

Unanimity versus majority. The most informative comparison is between the two committee filters with the same predictor. With the pool-vote predictor, unanimity filtering with CAL selection improves on the supervised ensemble by +0.53 points at 10% labels ($p = 0.032$) and +0.49 at 20% ($p = 0.067$), whereas majority filtering leaves it unchanged (-0.08 and -0.01 points, $p = 0.959$ and 0.906). Theorem 2 explains this: majority pseudo-labels are, by construction, the ensemble's own predictions, so for the ensemble they contain no new information ($g > 0$, no purification), while unanimous labels of a selected diverse committee are purer than the ensemble's average member. With a single-member predictor, both filters help (CAST-1: +0.56, $p = 0.025$; Maj-ST-1: +0.68, $p = 0.008$, at 10% labels), because any committee-derived label transfers knowledge of the other members into the single model.

Per-dataset behavior. Fig. 11 shows the per-dataset gains of CAST-V over Sup-Ens. They are positive on 23 of 34 datasets at 10% labels, largest on *vowel* (+5.65 points) and most negative on *tic-tac-toe* (-3.50 points). At 20% labels the largest gain is again on *vowel* (+7.90) and the largest loss is -1.32 points (*heart.c*). The gains grow with the number of classes (Spearman correlation 0.46 with C, $p = 0.006$, at 10% labels; 0.36, $p = 0.039$, at 20%), and the largest ones occur on problems with many confusable classes such as *vowel* and *letter*. This is consistent with the collision bonus of Proposition 1: with many classes, wrong members rarely agree, so unanimity removes errors more effectively.

E. ABLATION STUDY

Table 11 reports the ablations of CAST-1 (single-member predictor) on the first 20 datasets in alphabetical order, with three seeds and 10% labels. Besides test accuracy, it reports three diagnostics of the pseudo-labels: their precision on $\mathcal{U}$ in the selected round, the fraction of $\mathcal{U}$ that they cover, and the mean gap between the precision predicted by CAST and the observed precision of its filter (positive values mean over-confidence).

Committee selection. Using the whole pool as a unanimous committee (CAST-All) keeps the precision of the pseudo-labels high but covers far fewer points (PL frac. in Table 11) and changes accuracy by -0.47 points ($p = 0.084$), the largest loss among the variants that keep a diverse pool. This is the yield side of Proposition 2: beyond two or three members, unanimity costs more coverage than it gains in purity.

Correlation correction. Selecting with the independence

TABLE 9. Test accuracy (%, mean ± standard deviation over five seeds) with 20% labeled data. Best per dataset in bold.

Dataset	Sup-Best	Sup-Ens	ST-RF	Tri-Train	LabelSpread	Maj-ST-1	Maj-ST-V	CAST-1	CAST-V
anneal	84.7±2.9	84.8±1.6	**87.0**±2.0	**87.0**±2.8	74.6±3.6	84.7±3.3	85.2±2.2	83.9±2.3	85.1±1.0
balance.scale	87.0±1.4	85.7±1.8	76.7±3.2	80.0±2.5	83.6±1.4	**87.1**±1.5	86.0±1.7	87.0±1.4	86.2±1.7
breast.w	94.7±2.1	95.4±1.5	94.2±1.9	94.5±0.7	95.2±2.0	94.7±2.1	95.5±1.6	94.7±2.1	**95.7**±1.6
car	87.8±1.9	88.8±1.8	82.6±2.3	85.9±1.5	76.7±0.8	87.8±1.9	**88.9**±1.8	87.7±2.0	88.1±2.0
cmc	47.3±1.3	47.5±1.9	48.8±1.4	**48.9**±0.7	39.9±2.4	48.3±2.0	47.2±2.1	47.9±1.2	47.8±1.3
colic	75.7±8.2	79.1±3.7	78.9±5.4	80.4±5.7	64.6±10.4	76.7±8.8	77.8±1.8	77.0±8.5	**80.7**±4.1
column3C	81.3±3.8	82.8±2.6	**83.6**±3.6	81.8±2.7	75.6±4.0	81.5±3.3	82.3±1.9	81.8±3.3	82.6±3.4
credit.a	85.8±3.0	86.7±1.0	87.2±2.1	**88.4**±2.2	70.4±4.1	86.2±3.0	86.1±1.0	86.0±1.3	86.6±1.4
credit.g	72.2±2.2	**73.1**±1.7	70.6±0.7	71.4±1.4	68.8±1.5	71.7±1.6	72.3±1.8	71.2±2.3	71.8±1.2
cylinder.bands	65.9±6.3	**73.2**±2.4	67.1±2.9	69.3±4.2	68.1±1.9	65.6±6.2	71.0±5.0	67.3±6.6	72.1±3.1
dermatology	96.1±1.8	95.7±1.2	96.1±1.6	96.7±1.0	88.0±3.8	96.5±0.8	96.7±1.2	97.0±0.8	**97.6**±0.4
diabetes	72.6±2.6	**73.6**±3.0	72.4±0.9	73.0±3.0	68.9±3.3	72.6±2.6	73.2±2.7	72.7±2.6	73.5±1.6
ecoli	84.1±1.3	85.1±1.4	79.8±5.8	82.2±4.5	85.4±3.2	84.4±1.6	**86.6**±3.2	83.4±1.5	85.1±0.9
heart.c	79.5±1.3	**82.4**±1.6	78.7±3.3	79.7±5.2	78.2±2.7	80.3±2.4	81.8±2.7	80.5±1.3	81.1±2.4
hypothyroid	98.7±1.1	95.2±0.7	96.0±0.9	96.6±1.1	90.4±1.5	**99.0**±0.6	95.2±0.7	98.6±1.1	96.0±1.0
ionosphere	89.3±2.9	91.4±2.1	91.8±3.3	91.1±3.7	76.1±8.2	89.3±2.9	**92.5**±1.5	90.5±3.7	**92.5**±1.2
kr.vs.kp	95.0±0.2	**95.7**±0.5	94.6±1.6	95.5±0.7	78.9±0.9	95.4±0.7	**95.7**±0.2	94.8±1.1	95.3±0.9
letter	66.8±1.4	66.7±2.1	64.7±3.1	67.3±1.5	60.5±2.0	66.9±2.5	66.8±2.1	**71.1**±1.7	69.7±0.8
mushroom	99.5±0.6	**99.8**±0.1	99.6±0.4	99.6±0.4	99.5±0.6	99.6±0.6	99.7±0.2	99.5±0.6	**99.8**±0.1
nursery	89.3±1.7	**89.9**±1.2	87.8±1.4	88.2±1.3	72.5±1.4	89.0±1.2	89.6±1.7	88.8±1.2	89.8±1.2
optdigits	93.8±0.6	94.6±0.7	91.5±0.5	93.8±0.6	95.2±1.0	94.6±1.1	95.0±0.8	95.0±1.1	**95.9**±0.6
page.blocks	96.3±0.1	96.6±0.6	95.4±1.0	96.1±0.7	95.1±1.1	96.5±0.5	96.6±0.3	96.5±0.3	**96.7**±0.6
pendigits	94.0±0.9	94.0±0.8	92.1±1.2	93.9±1.2	**95.6**±0.8	94.0±0.9	94.2±1.1	95.4±1.2	94.7±1.0
segment	94.8±0.3	94.7±1.1	94.0±1.1	94.2±1.2	89.7±1.8	94.9±0.2	94.5±1.1	**95.0**±0.5	94.1±1.0
sick	96.4±1.0	96.2±0.7	96.4±0.7	**96.8**±0.8	94.7±0.6	96.0±1.4	95.9±1.3	96.6±1.1	96.1±0.9
soybean	87.2±2.9	88.4±2.5	82.7±3.5	84.6±1.0	84.1±3.4	88.5±1.9	**89.1**±1.8	88.0±2.4	88.5±2.1
spambase	92.3±0.5	92.0±0.6	91.2±0.5	91.8±0.7	84.1±1.5	92.2±0.6	**92.5**±0.7	92.2±0.5	92.1±0.9
splice	90.7±1.3	93.5±1.2	73.5±3.4	92.2±1.1	63.1±1.6	93.4±1.6	**94.8**±0.3	92.7±1.2	94.8±0.8
tic-tac-toe	**91.1**±5.4	82.7±3.5	74.4±1.7	80.2±2.4	65.9±3.0	**91.1**±5.4	82.3±3.8	**91.1**±5.4	83.4±5.6
vehicle	**73.7**±2.2	72.1±1.8	68.2±2.9	69.6±1.7	64.8±2.7	72.5±2.3	72.0±1.9	72.9±2.2	71.8±2.1
vote	93.0±2.1	94.5±1.8	**96.0**±1.4	94.7±2.3	88.6±3.0	93.6±2.9	94.3±2.0	93.0±2.1	94.5±2.6
vowel	65.8±2.4	52.2±2.9	64.4±4.1	**68.4**±2.6	33.8±2.3	64.3±4.0	52.2±2.9	67.7±3.5	60.1±1.9
waveform5000	81.8±0.9	83.4±0.9	80.9±1.2	82.2±1.6	72.0±1.1	82.6±1.5	83.7±0.3	83.0±1.1	**84.0**±0.5
fashion-mnist	73.5±2.3	76.4±1.5	72.0±2.0	73.0±0.9	68.9±1.0	74.8±2.1	76.0±1.9	75.2±3.3	**76.5**±1.5
Avg. acc.	84.64	84.82	82.67	84.39	76.81	84.89	84.81	85.16	85.30
Avg. rank	5.57	3.76	6.47	5.03	7.97	4.84	3.63	4.56	3.16
#Best	2	7	3	5	1	3	7	3	9

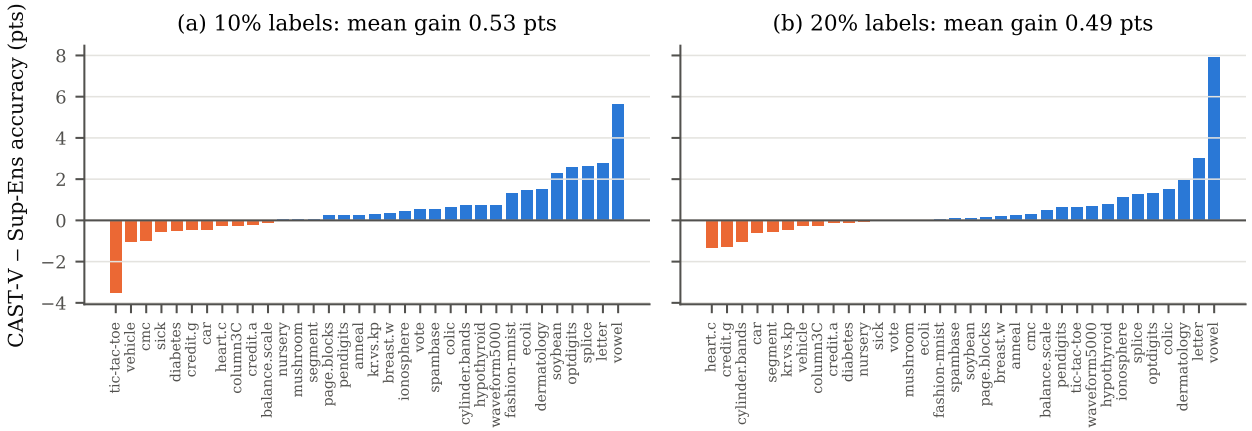


FIGURE 11. Per-dataset accuracy gain of CAST-V over the supervised ensemble Sup-Ens, which uses the same predictor without pseudo-labels.

law (CAST-Indep) yields slightly higher accuracy on this subset (+0.20, not significant) but is badly mis-calibrated: it over-estimates the precision of its own filter by +0.104 on average, against +0.007 for CAL. An over-confident predictor is harmless for selection when the ranking of committees is preserved, as here, but it would be misleading for any use of the predicted precision itself, such as the pseudo-label weights or a stopping criterion. Selecting with raw OOF counts (CAST-Emp) gives practically the same accuracy as CAL (+0.03).

Autocatalysis and safety. A single round (CAST-1R) and always using the last round (CAST-Unsafe) change accuracy by +0.10 and -0.11 points, respectively; neither difference is significant. The safe rule nevertheless matters for how much unlabeled data is trusted: CAST-Unsafe pseudo-labels a far larger fraction of $\mathcal{U}$ with lower precision, without any gain in

TABLE 10. Effect of pseudo-labels with the predictor held fixed: mean accuracy gain (percentage points) of each semi-supervised method over the supervised model with the same predictor, W/T/L over datasets, and Wilcoxon p-value.

	10% labels			20% labels		
Comparison	Gain	W/T/L	p	Gain	W/T/L	p
CAST-V vs Sup-Ens	+0.53	23/0/11	0.032	+0.49	19/3/12	0.067
Maj-ST-V vs Sup-Ens	-0.08	15/6/13	0.959	-0.01	15/3/16	0.906
CAST-1 vs Sup-Best	+0.56	21/2/11	0.025	+0.52	20/5/9	0.005
Maj-ST-1 vs Sup-Best	+0.68	21/6/7	0.008	+0.25	21/5/8	0.030
ST-RF vs Sup-Best	-2.96	10/1/23	0.003	-1.97	9/1/24	0.026
Tri-Train vs Sup-Best	-0.45	12/1/21	0.396	-0.25	18/1/15	0.791

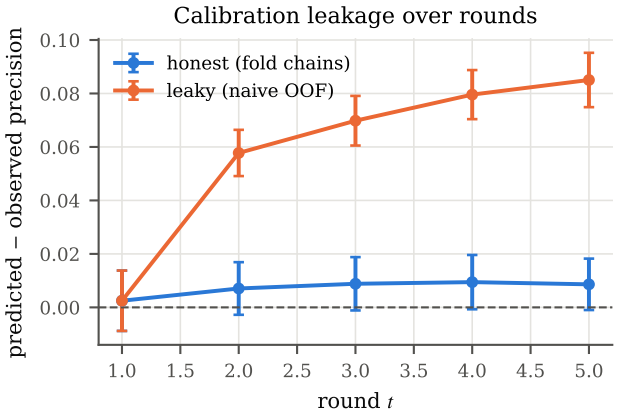


FIGURE 12. Calibration leakage. Mean difference between the predicted and the observed precision of the unanimity filter over rounds (10% labels, ablation subset, mean $\pm$ standard error). The naive OOF estimate drifts upward as pseudo-labels carry the validation labels back into training; fold-specific chains keep it close to zero.

accuracy.

Leakage. The leaky variant (CAST-Leaky) illustrates Section IV-G. Its calibration gap grows with the rounds and reaches +0.085 in round 5, against +0.009 for honest calibration (Fig. 12). Believing its pseudo-labels to be better than they are, it selects later rounds (mean $t^\star = 3.02$ versus 1.18) and trains on a much larger fraction of pseudo-labels. On these datasets its effect on accuracy is negligible (+0.07 points), but the inflated self-assessment is exactly the kind of error that a practitioner relying on validation scores would not detect.

Diversity. Replacing the diverse pool by seven bootstrap trees (CAST-Homog) lowers the precision of the pseudo-labels and changes the accuracy by -6.72 points. Part of the accuracy loss reflects the weaker single-tree predictor, but the lower pseudo-label precision at a higher coverage is the signature of the high error correlation of homogeneous members predicted by Corollary 1.

F. RUN TIME

Table 12 reports wall-clock training times per run; each run used a single CPU core, with two to three runs executing concurrently on a 2-core machine, so absolute times are pessimistic. CAST costs roughly an order of magnitude more than a supervised model, as predicted by (16), because of its calibration and fold chains; this is the price of leakage-free self-assessment. On the larger datasets of the benchmark ($n_L + n_U \geq 1400$) a CAST-V run takes about 23 seconds on average, which is acceptable for the label-scarce, moderate-size problems for which self-labeling is intended.

VIII. DISCUSSION

A. WHAT THE CHEMICAL VIEW EXPLAINS

The correspondence with asymmetric catalysis is not a metaphor laid over the algorithm afterwards; it is an identity (Theorem 2) with practical consequences. It explains in closed form three regularities that practitioners of disagreement-based learning know empirically. Diversity helps because it creates heterochiral pairs, and discarding them is what purifies the labels (Corollary 1). Majority rules dilute the effect because they let the meso complex react ($g > 0$). Self-training of a single model, or of a committee of near-identical models, cannot escape its own errors because amplification over rounds requires $M_{\text{eff}} > 1$ (Proposition 3). The bifurcation at $\chi = 1$ adds a warning that is easy to overlook: a highly susceptible self-trainer can converge to a confident labeling even with no information, and that labeling can be the mirror image of the truth, just as the sign of Soai's product is decided by chance.

B. PRACTICAL GUIDANCE

The results suggest four rules of thumb. (i) Build committees from different model families and measure their error correlation; members with ρ close to one add almost nothing. (ii) Prefer small unanimous committees of two or three members; the optimal size grows only when members are weak or correlated (Proposition 2). (iii) Weight pseudo-labels by the squared predicted excess rather than accepting them as clean labels. (iv) Never estimate the quality of a self-trained model with validation folds whose labels have influenced the pseudo-labels; use fold-specific chains or a separate hold-out.

C. LIMITATIONS AND THREATS TO VALIDITY

Approximation. CAL is exact for two members and in its limiting cases, but for larger correlated committees it summarizes the joint error structure by pairwise statistics, and its yield prediction is noticeably less accurate than its precision prediction (Table 4). When enough labeled data are available, the raw empirical frequency predicts precision better. Idealized dynamics. The mean-field map (12) assumes homogeneous members and a constant susceptibility; it explains qualitative behavior and should not be read as a quantitative model of any particular learner. Scope of the benchmark. Our experiments use classical learners on tabular data and a PCA representation of Fashion-MNIST, with datasets capped at 2,000 instances; deep semi-supervised methods with data augmentation were outside the scope of this study, and the gains of CAST in accuracy are modest. Design history. The final-predictor variant CAST-V was added after the initial

TABLE 11. Ablation study (10% labels, first 20 datasets, seeds 0–2). Δ: mean accuracy difference to CAST-1 in percentage points; p: Wilcoxon signed-rank test over datasets; PL prec.: precision of the pseudo-labels used by the final model; PL frac.: fraction of $\mathcal{U}$ pseudo-labeled; Calib. gap: mean predicted minus observed filter precision over all rounds.

Variant	Acc. (%)	Δ	p	Avg. rank	PL prec.	PL frac.	Calib. gap
CAST-1	80.12	–	–	4.03	0.868	0.19	+0.007
CAST-All	79.65	-0.47	0.084	5.12	0.854	0.07	-0.048
CAST-Indep	80.32	+0.20	0.460	3.62	0.843	0.20	+0.104
CAST-Emp	80.15	+0.03	0.687	4.00	0.856	0.19	+0.011
CAST-1R	80.23	+0.10	0.650	3.83	0.866	0.20	+0.002
CAST-Homog	73.41	-6.72	$<$0.001	7.30	0.832	0.24	-0.002
CAST-Leaky	80.19	+0.07	0.748	4.15	0.861	0.50	+0.059
CAST-Unsafe	80.02	-0.11	0.981	3.95	0.852	0.80	+0.007

TABLE 12. Training time per run in seconds (10% labels, all datasets and seeds).

Method	Mean	Median	Mean ($n_L + n_U \geq 1400$)
Sup-Best	1.04	0.88	1.24
Sup-Ens	0.38	0.35	0.49
ST-RF	1.23	1.06	1.57
Tri-Train	0.94	0.78	1.08
LabelSpread	0.19	0.18	0.30
Maj-ST-1	23.96	20.91	31.21
Maj-ST-V	31.82	28.81	40.11
CAST-1	12.94	10.36	17.10
CAST-V	18.15	16.35	23.00

design, once it was clear that the strongest supervised reference was an ensemble; it was validated on the pilot seeds before being run on the evaluation seeds, and we report both variants. Ablation subset. Because of compute limits, the ablation study uses the first 20 datasets in alphabetical order and three seeds. Cost. Leakage-free calibration multiplies training cost by roughly the number of folds.

D. FUTURE WORK

The analogy suggests several directions. Kagan's model has richer variants, including reservoir effects with partially active meso complexes and higher-order ML_n aggregates [18], [19]; their machine-learning counterparts are soft agreement rules and committees that vote in sub-groups. Soai's autocatalysis has been modeled with detailed kinetics [23], which could inspire better models of the susceptibility of deep networks to their own pseudo-labels. CAL could replace confidence thresholds inside modern deep SSL pipelines [24], [25], using several heads or augmentations as committee members. Finally, the leakage identified in Section IV-G is likely to affect other self-improving systems that validate themselves on data that shaped their own training signal.

IX. CONCLUSION

We have shown that agreement-filtered pseudo-labeling, one of the most common tools of semi-supervised learning, obeys the same law as the non-linear effects in asymmetric catalysis recognized by the 2026 Nobel Prize in Chemistry. For two classifiers the correspondence is exact: the purity of the labels they agree on is given by Kagan's ML_2 formula, with diversity playing the role of the meso complex and the treatment of disagreements playing the role of its reactivity. From this identity we derived a tanh law for independent committees, a Chiral Amplification Law for correlated multi-class committees, the optimal committee size, and a mean-field model of autocatalytic self-training with a symmetry-breaking bifurcation. We also identified a previously undescribed leak through which pseudo-labels inflate the validation estimates of self-training, and removed it with fold-specific pseudo-label chains.

On 20,400 real committees, the law predicts the precision of agreement filtering from labeled data alone with a mean absolute error of 0.069, and with 0.029 when its statistics are known, while the independence assumption common in the literature is systematically over-confident. The resulting algorithm, CAST, improves on its supervised counterpart and on classical self-labeling baselines in our benchmark, with gains that are consistent but modest; its most important practical property is that its self-assessment is honest. We hope that the chemical view of pseudo-labeling, in which classifiers are catalysts, agreements are homochiral complexes and self-training is autocatalysis, will prove as useful to machine learning as non-linear effects and autocatalysis have been to the synthesis of chiral molecules.

APPENDIX A PROOFS

PROOF OF THEOREM 1

With $C = 2$ every wrong prediction is the same label, so the filter accepts x iff all members are correct or all are wrong. By independence, $\Pr[\text{all correct}] = \prod_i a_i$ and $\Pr[\text{all wrong}] = \prod_i b_i$, which gives (4) and

$$\varepsilon_{\text{out}} = \frac{\prod_i a_i - \prod_i b_i}{\prod_i a_i + \prod_i b_i} = \frac{R-1}{R+1}, \qquad R = \prod_i \frac{a_i}{b_i}.$$

Since $a_i = (1+\varepsilon_i)/2$ and $b_i = (1-\varepsilon_i)/2$, $a_i/b_i = e^{2\,\operatorname{artanh}\varepsilon_i}$, so $R = e^{2u}$ with $u = \sum_i \operatorname{artanh}\varepsilon_i$, and $(e^{2u}-1)/(e^{2u}+1) = \tanh u$. For identical members, $(1+\varepsilon)^M + (1-\varepsilon)^M = (1-\varepsilon^2)^{M/2}\, 2\cosh(M \operatorname{artanh}\varepsilon)$, which yields the stated yield. ■

PROOF OF THEOREM 2

Let $c = \operatorname{Cov}(e_1, e_2) = \rho\sqrt{a_1 b_1 a_2 b_2}$. The error indicators and the correctness indicators $1 - e_i$ have the same covari-

ance, hence

$$P_{cc} = a_1 a_2 + c, \qquad P_{ww} = b_1 b_2 + c,$$
$$D = 1 - P_{cc} - P_{ww} = a_1 b_2 + a_2 b_1 - 2c,$$

which is (6). A disagreeing pair contains one correct and one wrong prediction; labeling it by a uniformly chosen member produces a correct label with probability $1/2$, so accepted disagreements contribute mass gD and zero net excess. The accepted mass is $(1-D)+gD$ and the net excess mass is

$$P_{cc} - P_{ww} = a_1 a_2 - b_1 b_2 = a_1 + a_2 - 1 = \bar{\varepsilon},$$

in which c cancels. Therefore

$$\varepsilon_{\text{out}} = \frac{\bar{\varepsilon}}{(1-D)+gD} = \frac{\bar{\varepsilon}}{(1-D)(1+g\beta)} = \bar{\varepsilon}\,\frac{1+\beta}{1+g\beta},$$

using $\beta = D/(1-D)$ and $1/(1-D) = 1+\beta$. ■

PROOF OF COROLLARY 1

(i) With $g=0$, $\varepsilon_{\text{out}} = \bar{\varepsilon}(1+\beta)$ and $\beta \geq 0$, with $\beta = 0$ iff $D = 0$. (ii) With $a_1 = a_2 = a$, $D = 2ab(1-\rho)$ and $2ab = (1-\varepsilon^2)/2$, so $1+\beta = 1/(1-D) = 1/[1-(1-\rho)(1-\varepsilon^2)/2]$, which is decreasing in ρ, equals 1 at $\rho = 1$ and $2/(1+\varepsilon^2)$ at $\rho = 0$. (iii) With $g = 1$ the factor is 1. ■

PROOF OF PROPOSITION 1

For two members, an accepted point is either a correct agreement (probability P_{cc}) or a wrong agreement, which requires both to be wrong and to choose the same wrong label (probability κP_{ww} by definition of κ). For M independent members erring uniformly, all M are wrong with probability $\prod_i b_i$ and then choose a common label with probability $(C-1)(C-1)^{-M}$, so $\pi/(1-\pi) = \prod_i (a_i/b_i)\,(C-1)^{M-1}$; taking logarithms gives (7). ■

PROOF OF PROPOSITION 2

The expression for Φ follows from Theorem 1. As $\varepsilon \to 0$, $\theta \sim \varepsilon$, $\tanh(M\theta) \sim M\varepsilon$, $\cosh(M\theta) \to 1$ and $(1-\varepsilon^2)^{M/2} \to 1$, so $\Phi(M)/\varepsilon^2 \to M^2 2^{1-M}$. The sequence $M^2 2^{1-M}$ takes the values $1, 2, 2.25, 2, 1.5625, \ldots$ for $M = 1, 2, 3, 4, 5, \ldots$ and is decreasing for $M \geq 3$, so it is maximized at $M = 3$. The threshold above which $M^\star = 2$ was located numerically on a grid of 10^4 values of $\varepsilon \in (0,1)$; on the same grid $M^\star \in \{2, 3\}$ for every ε. ■

PROOF OF PROPOSITION 3

Let $\phi(x) = \tanh(M_{\text{eff}} \operatorname{artanh} x)$ on $(-1,1)$. Writing $u = \operatorname{artanh} x$, $\phi'(x) = M_{\text{eff}} \cosh^2 u / \cosh^2(M_{\text{eff}} u)$, and

$$\frac{d}{du}\frac{\cosh(M_{\text{eff}} u)}{\cosh u} = \frac{\cosh(M_{\text{eff}} u)}{\cosh u} \times \left(M_{\text{eff}} \tanh(M_{\text{eff}} u) - \tanh u\right) > 0$$

for $u > 0$ and $M_{\text{eff}} > 1$. Hence ϕ' is decreasing on $[0,1)$, i.e. ϕ is strictly concave there; ϕ is odd, $\phi(0) = 0$, $\phi'(0) = M_{\text{eff}}$, and $\phi(x) \geq x$ on $[0,1)$ because $M_{\text{eff}} u \geq u$. Let $f(x) = (1-s)\varepsilon_0 + s\phi(x)$.

Case $\varepsilon_0 > 0$. $f - \mathrm{id}$ is concave on $[0,1)$, positive at 0 and tends to $(1-s)\varepsilon_0 + s - 1 < 0$ as $x \to 1$, so it has exactly one zero $\varepsilon^\star$. Since $f(\varepsilon_0) - \varepsilon_0 = s(\phi(\varepsilon_0) - \varepsilon_0) \geq 0$ and f is increasing, the iterates starting at ε_0 increase monotonically and remain below $\varepsilon^\star$, hence converge to it; $\varepsilon^\star = \varepsilon_0$ iff $\phi(\varepsilon_0) = \varepsilon_0$ iff $M_{\text{eff}} = 1$. Monotonicity of $\varepsilon^\star$ in M_{eff} and s follows because f is pointwise increasing in both on $[\varepsilon_0, 1)$.

Case $\varepsilon_0 = 0$. $f = s\phi$ is odd with $f'(0) = \chi$. If $\chi \leq 1$, strict concavity on $(0,1)$ gives $f(x) < \chi x \leq x$ for $x > 0$ (for $M_{\text{eff}} = 1$, $f(x) = sx < x$), so 0 is the only fixed point. If $\chi > 1$, $f(x) > x$ for small $x > 0$ and $f(x) \to s < 1$ as $x \to 1$, so by concavity there is exactly one positive fixed point $\varepsilon^\star$, at which $f'(\varepsilon^\star) < 1$ (stable), and by oddness $-\varepsilon^\star$ is also a stable fixed point while 0, with $f'(0) = \chi > 1$, is unstable. ■

PROOF OF PROPOSITION 4

By induction on t: $\mathcal{P}_0^{(f)} = \emptyset$; if $\mathcal{P}_{t-1}^{(f)}$ is a function of $(X_{\mathcal{L}}, y_{\mathcal{L}\setminus V_f}, X_{\mathcal{U}})$, then so is every model trained on $(\mathcal{L} \setminus V_f) \cup \mathcal{P}_{t-1}^{(f)}$, the committee predictions on $\mathcal{U}$, and hence $\mathcal{P}_t^{(f)}$. The argument is conditional on the committees $S_1, \ldots, S_{t-1}$: these are chosen from statistics pooled over all folds and therefore depend weakly on y_{V_f} through a discrete choice among 120 candidates. Given the committees, the models evaluated on V_f are independent of y_{V_f}, which is the defining property of a hold-out estimate. The residual dependence through the committee choice is the usual, bounded selection optimism of cross-validated model choice [42], not a transfer of individual labels. ■

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